Cisco CCNA 200-301 - The Complete Guide To Getting Certified

David Corzo

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Chapter 1

Host to host communications

1.0.1 A very basic introduction to networking

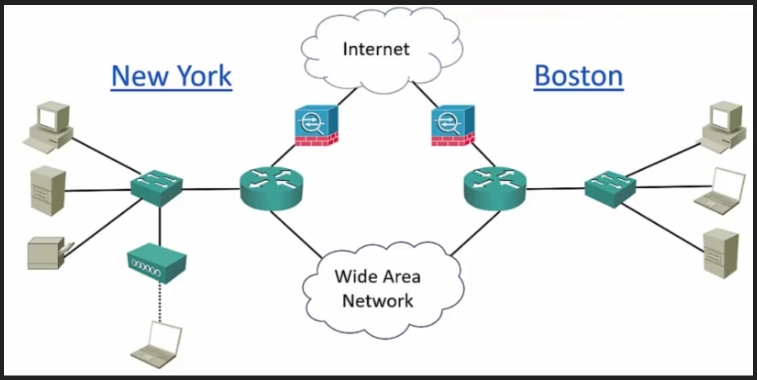
Let's say we have an office in NY. In that office there are end-hosts, such as PCs, server, printers, etc. All of the end hosts need to communicate with each other, for this we introduce a switch, this switch is going to be connected to each of the end-hosts using ethernet cables. A switch is a device with a lot of ports - which are basically slots to connect ethernet cables to - this allows the device to be connected to each other through the switch. The switch is what allows connectivity on my *local area network*. Let's say we add another end-host, a laptop, which is connected wirelessly to the same switch, the laptop will be connected via WIFI to a *wireless access point* and the wireless access point is connected to the switch via an ethernet cable. Now that all the end-hosts are able to communicate, we have built a *local area network*, which means a network that connects devices within the same local area, such as an office or a university campus.

Now that we have built a local area network, we also want these end-hosts to be able to communicate with other devices out on the internet, not just in my local area network. For this we introduce another component, the *router*, which is a device capable of making advanced routing decisions to route traffic between different areas of your network.

Next, now that we have connected the end-hosts to the local are network, and the local area network to the internet via a router. The internet is a very dangerous place, hackers can attack us and other things that we don't want can happen. For security reasons we install a *firewall*, which secures the different parts of our network from each other.

In a larger company we would not just have end-hosts connected to switches connected to routers connected to firewalls connected to the internet. We would have similar local area networks in many other places, for example, we can say that we have a very similar local area network in Boston. The same devices and the same connections. Lets say we want the NY and Boston office to communicate directly without using the internet. We can use a *dedicated connection* between my two routers on both sites for communications to be direct from the NY router to the Boston router, this is called a *wide area network*.

All this is expressed in the next, *network topology diagram*.



Some other caracteristics of networks are:

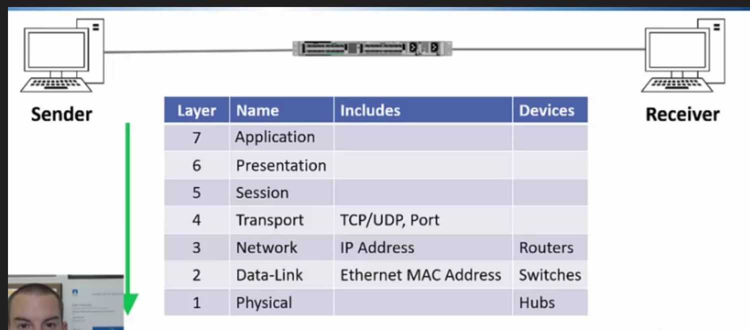
- Topology: how the devices are connected to the each other.
- Speed: the faster, the more cost.
- Cost: the monetary cost, effected by the type of devices and technologies used.
- Security: all the security features inside and outside the devices.
- Availability: making sure that our network always stays up. Mission critical components are typically duplicated, that way we can have an alternative if one were to fail.
- Scalability: the topology design lends itself to grow as needed.
- Reliability: similar to availability, making sure the network is going to continue to work.

1.1 The OSI Reference model

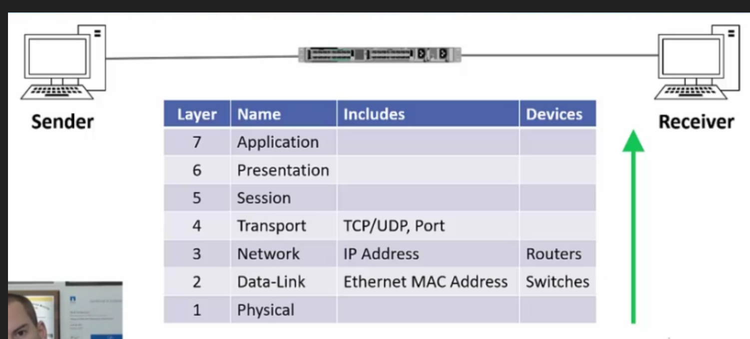
- The OSI model stands for Open Systems Interconnect model.
- It is standardized by the Organization for Standardization (ISO).
- It is a general purpose network that standardizes how computers communicate with one another over a network.
- The OSI model is conceptual it is not a physical thing, device, or protocol. It is a concept for humans only, to be able to understand.
- The 7 layer approach to data transmission divides the operations into specific related groups of actions at each layer. A layer serves the layer above it and is served by the layer below it.

1.1.1 Example

An example is how you send an email. The sender (the person sending the email) and the receiver (the person receiving the email). In the layer 7 - the application layer - which is the layer containing the “from:” or the “to:” field, this is collected at an application layer. Next the data will be encapsulated by the layer 6 - presentation layer - which deals with the format the data is being handled with. Then the layer 6 information is encapsulated into the layer 5 - session layer - information. That then gets encapsulated to the 4th layer - the transport layer - which is the layer that specifies what communcation will be used - TCP, UDP, Port number - to prepare it for the next layer. That gets encapsulated into layer 3 - the network layer - which includes the source and destination IP address, a device that operates at layer 3 is the routers. The next the packet is almost being finished, and passed on to the layer 2 - the data link layer - which specifies the mac address - in case we are sending this over an ethernet network -, devices that operate at this layer are switches. Finally the encapsulated and consolidated packet is ready to be sent, this is done in layer 1 - which is the Physical layer - which literally means sending the packet through the wire, a network device that used to be used in this layer are hubs - which are not used anymore -. The email has been sent. The packets travel through the wires and devices to the receiver and the receiver receives packets in the form of electrical signals.



The receiver will process the information being received from layer 1 to layer 7, which means that the receiver will grab the data from the first layer and decapsulate it until it has layer 7 data. It comes in at the physical layer, then it it will pass to the data link layer which is 2nd layer, it will perform checks to ensure that the the destination mac address is really for the receiver, if it is not, the receiver will just discard the packet. Next it will look at the layer 3 - which is the network layer - it will check the destination ip address and check if the packet is for it, otherwise it will discard the packet. Then we carry on with the next layer, layer 4 - the transport layer - it will check the port number in which it was received and some other validations. It will continue through the session layer - layer 5 -, the presentation layer - layer 6 -, and the application layer - layer 7 -.



Layer 6 and 7 - application and presentation - are considered the upper level, typically used by developers not network engineers. Usually network engineers start to become interested from layer 4 downward.

1.1.2 Benefits of the OSI model

- Engineers don't need to design technologies to work end to end from top to bottom of the model. Instead they can just focus on their layer of expertise, and make sure they comply with the standards for the layers above and below.
- This leads to open standards and multi-vendor interoperability.
- For example: if you're an application developer, you can just focus on the top three layers, the lower layers are the domain of network engineers. You don't need to understand thzem, you just make sure you are complying with the standards.
- Troubleshooting is easier because you can analyse a problem in a logical fashion layer by layer.
- It is difficult to convey how important the OSI model is. The OSI model is used to describe problems, solutions, technologies, etcetera. Such as if the cable was disconnected you would say 'it was a layer 1 problem', if you find that the problem was that the user specified a wrong destination - you say - 'it was a layer 7 problem, etcetera.

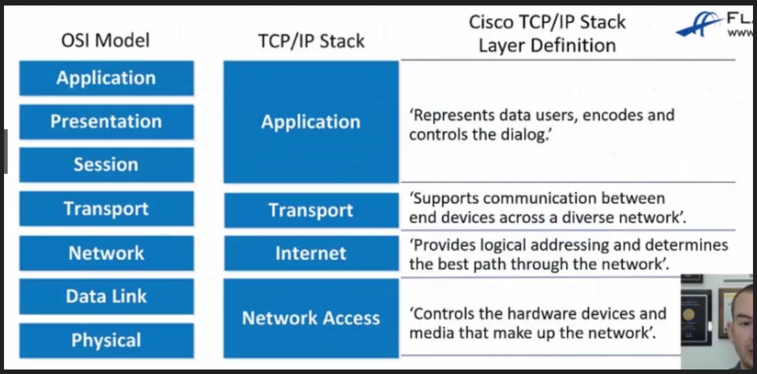
1.1.3 OSI Acronyms

- The classic: Please Do Not Throw Sausage Pizza Away.
- Network Relevant: Please Don't Need Those Stupid Packets Anyway.
- Us relevant: Please Do Not Teach Students Pointless Acronyms.
- Useful: Please Do Not Take Sales People's Advice.
- Favorite: Please Do Not Touch Superman's Private Area.

1.2 TCP/IP Stack

- TCP/IP was developed during the 1960s by the US department of defense's Advanced Research Projects Agency (ARPA).
- It is a protocol stack which consists of multiple protocols including TCP (Transmission Control Protocol) and IP (Internet Protocol). It is not a single protocol, it is a protocol stack.
- An analogy to understand what a protocol is is when diplomats meet each other, there is a certain way they are expected to behave and to communicate with each other, in computer networking protocols basically mean the same thing. You have two hosts and they are going to communicate with each other, there has to be a protocol that describes that communication is going to be and how is it going to behave.
- It is the main protocol stack used in computer operations today.
  - There used to be many other protocols such as Ipxspx and apple talk. But today TCP/IP is pretty much standard, other protocols are pretty much dead.
- Whereas the OSI reference model is conceptual, the TCP/IP stack is actually used to transfer data in production networks.
- TCP/IP is also layered but does not use all the OSI layers, though the layers are equivalent in operation and function.
  - It does actually use all of the layers, but in the documentations only 4 are layed out.

1.2.1 Comparing the OSI model with the TCP/IP



Notice that the TCP/IP does cover all the layers of the OSI model, however it sometimes uses the layers together. The application layer - in the TCP/IP - maps to the OSI model's layer 5,6, and 7 - session, presentation, and application -. Below that we have the transport layer - in the TCP/IP stack - which notice maps directly to the transport layer in the OSI model. The internet layer in the TCP/IP is equivalent to the network layer in the OSI model. Last we have the network access in the TCP/IP stack that maps to the OSI layer 1 and 2 - physical and data-link-.

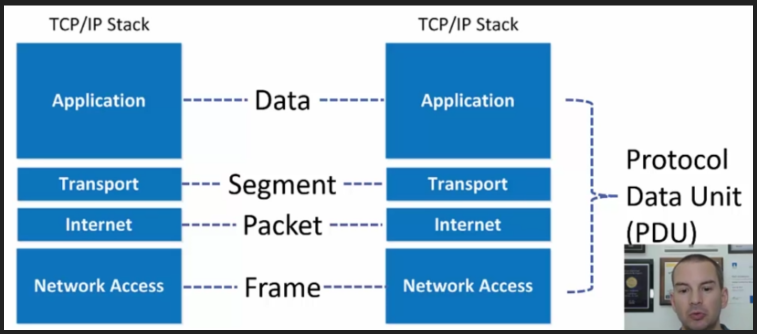
- While both the OSI model and the TCP/IP stack are very important in networking. We are going to typically be referencing more the OSI model than the TCP/IP stack.

1.2.2 Hosts communication terminology

When two hosts communicate with each other, they talk using something called protocol data unit (PDU), which is the entire communication that happens on the OSI model (sender from layer 7 to layer 1 and receiver from layer 1 back to layer 7).

The data exchanged varies in terminology according to which layer of the TCP/IP stack is being talked about.

- The application layer communicates using data to the application layer in the other host.
- The transport layer communicates using segments to the transport layer in the other host.
- The internet layer communicates using packets to the internet layer in the other host.
- The network access layer communicates using frames to the network access layer of the other host.



In the networking world the OSI model is more commonly referenced than the TCP/IP stack in terms of layers - which means that it is more common to refer to the layers of the OSI model than the layers of the TCP/IP -. However when we are talking about host communications terminology, it is more common to speak of the TCP/IP communications terminology - which means that it is more common to use 'data' if you are refering to the layer 5,6, and 7 of the OSI model, layer 3 call it a packet, layer 4 call it a segment-. Another very important thing is that networking people usually refer to the PDU as the name packet, the correct name is a PDU not a packet since the packet is used for layer 2 (internet layer) communication, but people typically use the term packet and PDU interchangeably, while technically wrong, but in day-to-day communcations when refering to host communications it is not uncommon to call it packet meaning the entire stack.

1.3 The upper OSI layers

- The networking engineers do not typically work directly with the upper 3 layers of the OSI model, but we still need to know what they do.
- They are more relevant to application developers.
- In this lecture I will primarily be giving you the Cisco definitions of the layers.
- Information included in the upper layers would include the message body and subject line in an email message for example.

1.3.1 Layer 7 - the application layer

- The application layer provides network services to the applications of the end user.
- If differs from the other layers in that it does not provide services to any other OSI layer.
- The application layer establishes the availability of intended communications partners.
  - Intended communcation partners are the hosts that the current hosts is trying to communicate with.
- It then synchronizes and establishes agreement on procedures for error recovery and control of data integrity.
  - Data integrity is verifying that data has not been corrupted or damaged in transit.

1.3.2 Layer 6 - the presentation layer

- The presentation layer ensures that the information that is sent at the application layer of one system is readable by the application layer of another system.
- The presentation layer can translate among multiple data formats using a common format (eg computers with different encoding schemes)

1.3.3 Layer 5 - session layer

- The session layer establishes, manages, and terminates sessions between two communicating hosts.
- THe session layer also synchronizes dialog between the presentation layers of the two hosts and manages their data exchange.
- For example, web servers have many users, so there are many communication processes open (session) at anygiven time to track. The session layer keeps track of all of the opened sessions.
- It also offers efficient data transfers, CoS (Class of Service), and exception reporting of upper layer problems.

1.4 The lower OSI layers

- Whereas network engineers are not particularly interested in the upper OSI layers, we are very concerned with the lower 4 layers of the OSI model. These are incredibly important and are almost like the daily routine to network engineer.
- Each of these layers have their own dedicated section later and you will learn much more detailed information about them throughout the course.

1.4.1 Layer 4 - transport layer

- The main characteristics of the transport layer are wether TCP or UDP transport is used, and the port number.
- We haven't really seen what is TCP and UDP, for now you can think of it as modes of transportation where each one has different advantages. The TCP is very reliable, therefore if priority is reliability we use TCP. If speed is more important than reliability - like for voice of video traffic - then we use UDP instead.
- We haven't seen what ports are, but for now you just need to know that different processes occur in these ports, for example the 80th port is used for http web traffic and port 25 is used for email.

- Definition:
  - The transport layer defines services to segment, transfer, and reassemble the data for individual communications between the end devices.
  - It breaks down large files into smaller segments that are less likely to incur transmission problems.

1.4.2 Layer 3 - network layer

- The most important information at the network layer is the source and destination IP address. Other information is also carried, but the main is the source and destination of IPs.
- Routers operate at layer 3.
- Definition:
  - The network layer provides connectivity and path selection between two host systems that may be located on geographically separated networks.
  - The network layer is the layer that manages the connectivity of hosts by providing logical addressing (IP addressing is our logical addressing).

1.4.3 Layer 3 - the data-link layer

- The most important information at the data-link layer is the source and destination layer 2 address. Again - just like in layer 3 and 4 - there is also other information but the main one is the source and destination layer address.
- For example the source and destination MAC address if ethernet is the layer 2 technology. Different layer 2 technologies use different formats for their addressing, for example, old legacy frame relay uses DLCI or DCE numbers for the addressing. Today the typical address used at this layer is the MAC address.
- Switches operate at layer 2.
- Definition:
  - The data link layer defines how data is formated for transmission and how access to physical media is controlled.
  - It also typically includes error detection and correction to ensure a reliable delivery of data.

1.4.4 Layer 1 - the physical layer

- The physical layer concerns literally the physical components of the network, for example the cables being used.
- Definition:
  - The physical link enables bit transmission between end devices.
  - It defines specifications needed for activating, maintaining, and deactivating the physical link between end devices.
  - For example, voltage levels, physical data rates, maximum transmission distances, physical connectors, etcetera.

Chapter 2

The cisco IOS Operating System

2.1 Introduction

By now we all want to configure the cisco devices such as routers and switches. Cisco devices do include GUIs - which means graphical user interface - but in the real world, everybody just uses the command line interface (CLI) because it is much quicker than using the GUIs. The OS used in enterprise grid cisco devices is IOS, in this section we'll learn how to connect to the devices and the use of the command line. A disclaimer is that the IOS operating system is not the only one, there are others.

2.2 Cisco operating systems

Cisco history is not going to be examined on the CCNA exam. Neil includes this just because of context. Cisco IOS and the other OS's work basically the same way in terms of management and administrative point of view, the operating systems don't differ that much in terms of how you use it, maybe on how they are actually implemented but not really on how you use it. By learning cisco IOS you are really learning all of the others at once.

2.2.1 A short history of cisco operating systems

- Most people think of cisco as primarily a routing and switching company, but they actually started out with just routers in 1984.
- IOS was the original operating system that they used on routers - the same OS they use today (with some optimizations).
- IOS is the operating system that has been used on cisco routers since their inception.
- Cisco Catalyst switches evolved from the acquisition of Crescendo company in 1993 - cisco bought the company Crescendo- and started manufacturing switches of which the first one was Catalyst.
- The original cisco switch operating system was CatOS which has not been deprecated.
- Cisco firewalls evolved from the acquisition of the company Network Translation's PIX firewall that uses the Finesse operating system in 1995.
- Cisco switches and firewalls (now the ASA firewall) were ported over the IOS operating system over the following years.
- Cisco standardized on IOS for all the network infrastructure devices.

There are some other operating systems in some newer router and switch platform. IOS runs on the majority of routers and switches. Some of these other operating systems are:

- The Cisco Nexus and MDS data center switch product lines run on NX-OS.
- The IOS-XR operating system runs on the service provider NCS, CRS, ASR9000, and XR12000 series routers (these are the higher end routers).
- IOS-XE runs on the ASR1000 series service provider routers (1K service providers).
- The command line interfaces for teh other operating systems are nearly identical to IOS.

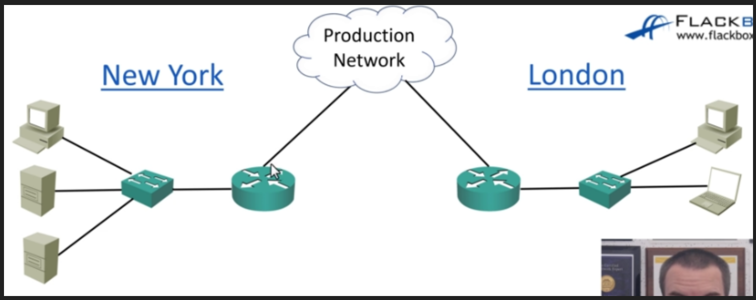
At first all of these names might seem intimidating but fear not, all of the OS's run nearly identical commands, if you can use one, you can practically use them all. There are different OS's because they are implemented with different functionality, for example, IOS has a monolithic kernel - which means that if one process crashes it crashes the entire router - they are all different under the hood. Other OS's have different kernels to permit that if one crashes the other processes in the router are not affected, only that process is crashed. This is does not mean that IOS is bad or unreliable, it simply means that it works differently given different situations.

2.3 Connecting to a cisco device over the network

- The lab exercises in this course use Cisco Packet Tracer simulation software on your PC.
- This lecture shows how to connect to a real router or switch over the network with Putty - which a software to connect to routers - you are not going to use Putty in the labs because everything is done in the Packet Tracer simulation software.
- You do not need to install or use Putty to do the course lab exercises.

2.3.1 Connecting over the network

In you day-to-day as a network engineer you are typically not going to be connecting stuff, instead you are going to be in your desk 'working remotely', you are going to be connecting to that device over the network. So the following example: Let's say that you are in a London office and want to establish a connection in the NY office. What you are going to do is use software on your PC, connect to the router over the network in NY, get to the command line of device you want to work on, and get on with your job.

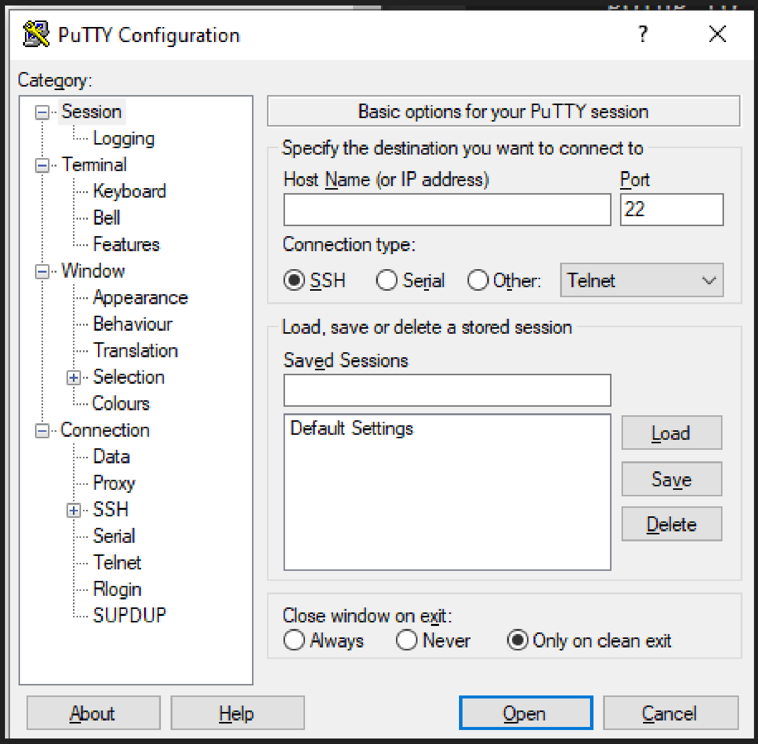


- To get to the CLI for day-to-day management of a Cisco device you will use Secure Shell (SSH) to connect to it's management IP addresses over the network.
- There is also another protocol called Telnet which is also supported but not recommended because it is insecure. Commands in Telnet are un-encrypted, so anybody listening in to your traffic can get ahold of data, including username and password. Telnet is never used in a real setting for the potential vulnerabilities. In reality SSH is always used, Telnet is almost always used only on a lab environment to learn.
- In enterprise networks, secure login will typically be enforced through intergration with a centralized AAA (Authentication, Authorization, and Accounting) server.

– Basically the AAA server is a solution to many people loggin in. In a network, you can probably have many devices with username and passwords that must be verified to be granted access to network actions. We don't want each device to have the username and passwords of all the others so we centralize that into one server that stores all the user name and passwords that has access to the central database of usernames and passwords.

2.3.2 Installing Putty

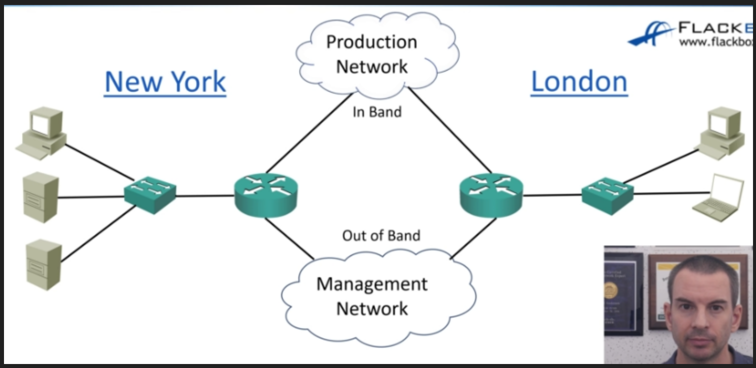
Installing Putty is basically just searching on Google and accepting all the defaults once the executable has been downloaded. When you launch the window you will be faced with a window like this:



The host name refers to the host that I am intending to connect to, it takes as an input an IP address. Then you log in through the command line, and type your password. This is how you connect to a cisco device, very simple using Putty.

2.3.3 Out of band management

To continue the example in which we were connecting in London to the NY office, we are essentially using the production network. But most large companies use a separate management network, this is done as a backup path in case the production network is down, it is also more secure to have a separate dedicated path. The production network is the in band network, which means you are using the same cables that your normal users (your staff) are using to connect to other devices. If you connect to the NY office through the management network you are using and out of band network, which means you are not using the same network as the normal staff. Trivial difference but important distinction.



2.4 Making the initial connection to a cisco device

- Cisco devices do not usually have a default IP address, so we need to set one up before we can connect to it over the network, but how do we connect to the device in the first place if we don't have an IP?

- We need a way to connect to the device to do the initial configuration including adding IP addresses. This is where the console connection comes in. With a console connection we can connect to the device at a lower level below IP to get to the initial command line to do the initial configuration including adding an IP and finally be able to connect to it over the network.

For this initial configuration, we need a Console Cable (DB9 to RJ45) in order to connect to the router via a wire and send the configurations. Simply connect the cable to the console port on the router or switch and use the cable - which usually comes with your device.

Figure 2.1: Console Cable DB9 to RJ45

The console cable has two ends, one is the DB9 which is the thicker end, it is called that because it has 9 pins need to be connected to that end.

Figure 2.2: DB9 end

The other end is very similar to an ethernet cable. However, THIS IS NOT AN ETHERNET CABLE. It is a RJ45, it is not using ethernet and it doesn't require IP addresses, this only grants you low level access to the command line.

Figure 2.3: RJ45

When you connect to make the initial configurations to the router using the console cable, the RJ45 - the one that looks like an ethernet connection but is not - end of the cable is the one that is going to be connected to the switch. The other end - the serial connector - plugs in to the computer. This is a problem for computers that don't include the DB9 connector, for this you can buy an adaptor from DB9 to USB.

2.4.1 Newer console cables

With newer cisco devices the cable included comes already with a USB end for the computer since most computers don't come with DB9 cable support. So the newer console cables come with some modifications, not only the USB end for the computer but they actually removed the RJ45 ethernet looking but not ethernet end to be instead a microusb end.

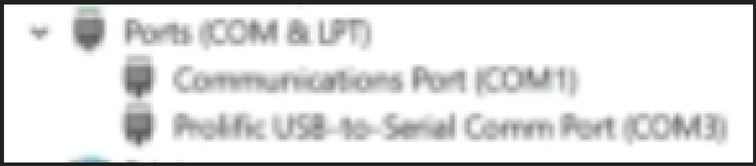
Figure 2.4: USB to microusb

2.4.2 Connecting via console cable to a real router

First identify the console port on the router and connect the appropriate end (microusb or RJ45) to the router, the other end to your computer (USB or DB9). Launch putty and click that the connection type is going to be 'Serial'. Now you are going to have to specify your COM port correctly, if you don't know which one it is follow the following steps (windows):

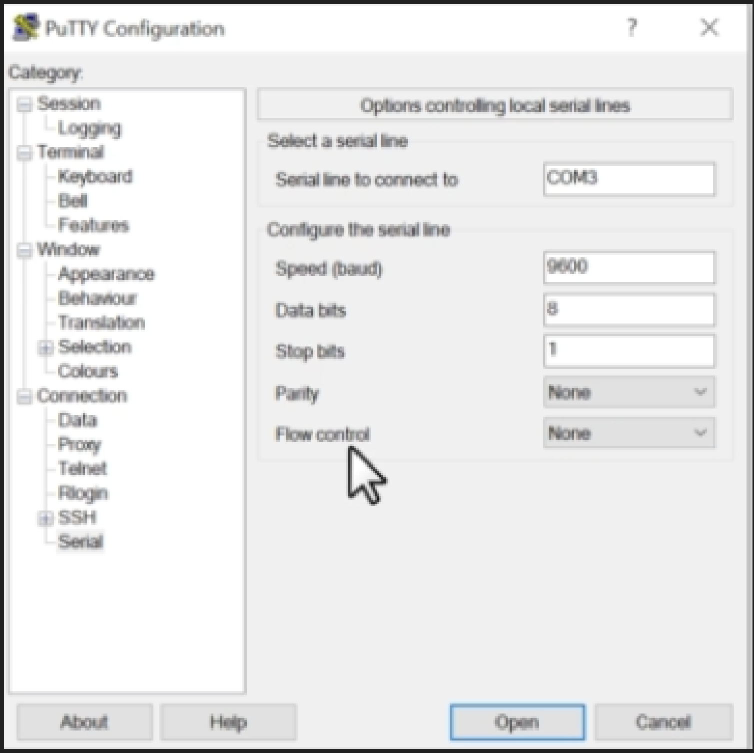
- Make sure that you have installed the driver software for the console cable in order for the COM port to be enabled.

- Open device manager and look in ’Ports (COM & LTP)’.



- In this case it is COM3, change in putty the COM¡number¿ port specified in the device manager.

Then you need to select the speed, specified in the ’Serial’ tab, and set flow control to None.



Return to the ’Session’ tab and click on ’Open’. Now turn on your router, you will start to get output on the screen. The next thing is that the router will boot up, this will take some time so give it some time.

2.4.3 Console connection troubleshooting

- When we buy a new device we saw how we can make the initial configurations including adding the IP addresses in order to connect to it over the network.

- As well as for initial configuration, the console port - and cable - can be used if the device’s IP addresses become unresponsive.

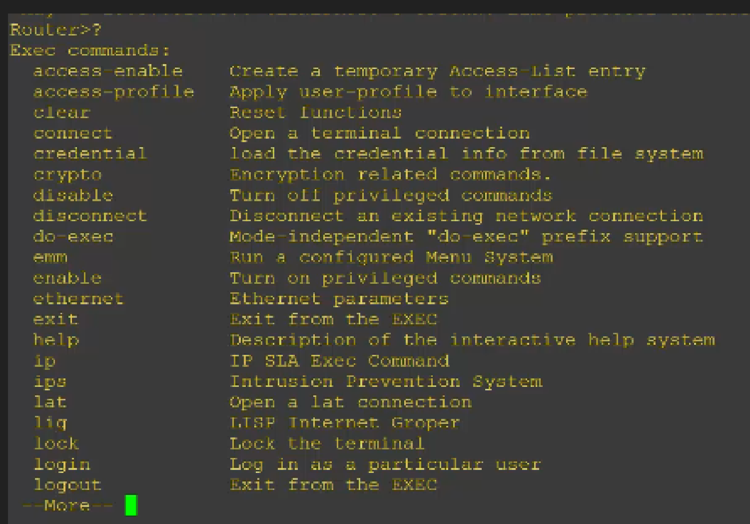
- It can also be used to troubleshoot the bootup process. You can view the device booting up from a console connection but this is not possible with SSH because the system must have booted already before the IP addresses will be live.

- Another case is when a device just appears to be completely unresponsive but you can see that the device is on, that indicates that the device is probably failing to boot up. Again trying to connect to it over the network will not work, you need to take out your console cable and troubleshoot it. It will not work by connecting to it because the IP is only live until it has completed the bootup process, therefore you cannot connect it through the network, but you can connect to it over the cable and power the device on and you can watch if any problems ensue.

2.5 Navigating the cisco IOS Operating System Part 1

2.5.1 Booting up, and prompt initial configurations

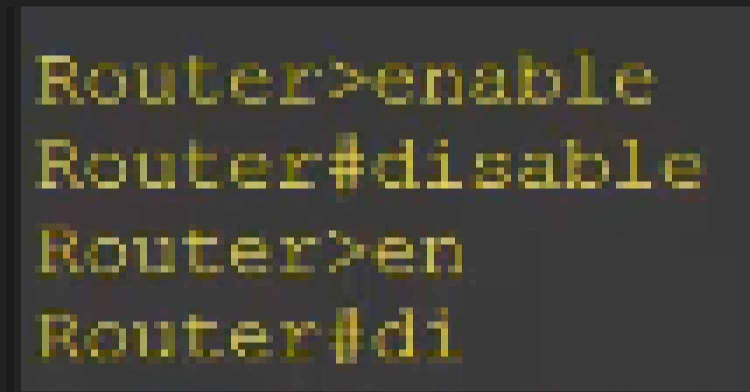
Once you buy a router and connect the console cable to it, output will be displayed on the screen of the PC you connecting the other end of the console cable to. This means that the router is booting up. Once it has finished booting up a prompt will appear and ask you if "Would you like to enter the initial configuration dialogue? [yes/no]: ". If I say yes, it will take me to a command line driven wizard which would help me do the initial setup. However this option is not usually used, so we will say no. Once you say no, another prompt will appear "Would you like to terminate autoinstall? [yes]: " type yes. Once you have said yes to the previous prompt the command line will take you to the user exec prompt, however this command line environment is very limited in the number of commands we can process. If you want to see which commands are supported in the exec prompt type "?". This will print some of the commands, the 'More' word at the bottom signals that there are more commands, to see the rest of the commands we can hit the enter key and it will display the new commands line by line, by using the space bar you can display the commands page by page.



Now, as you continue to type commands and output is printed to the screen, some previously typed lines are going to be erased, to save them you hit on Putty reconfiguration, go to the window tab, in the attribute called "Lines of scrollback: " set it to a large number such as 2,000.

2.5.2 Switching from user exec prompt to privilaged exec mode

As we said the exec prompt is very limited and we can't do much in this prompt, so we are going to switch to another prompt called the privileged exec mode, to go to a privileged exec mode simply type "enable" or it's abbreviation "en". Then the command prompt changes, now instead of each line being something like "Router>" you have something like "Router#". To be switch from privileged exec mode to normal exec user mode simply type "Disable" or it's abbreviation "disa".



2.5.3 Ambiguous abbreviations

Abbreviations for command need to map uniquely to one command. For example we can abbreviate the "enable" command to just "en" because it is the only command that starts with "en" therefore that abbreviation is not ambiguous. With the "disable" command we cannot abbreviate as "di" because it is not the only command that starts with "di" other commands such as "dir" and "disconnect" also starts with "di", therefore to make it unique we not only need to add an "s" to "di" but also an "a" to make it "disa". To see if an abbreviation is ambigous you can type the abbreviation followed by a question mark. Such as: `Router# di?`. It is important that when you are checking this the question mark is included after the proposed abbreviation without a space because a space and then question mark is something else.

2.5.4 The show command, tab completion and errors

The "show" command - or it's abbreviation "sh" are used to be able to see information inside the kernel. When you type in abbreviation followed by a space followed by a question mark, the kernel will return all the possible keywords that are available for that particular command, this is known as context sensitive help - which means which combinations of commands we can use together. If we type "sh ?" it will print all keywords that can be used with the "sh" command before it. If we say something like "sh aaa ?" which are both commands, are asking the kernel to tell us which other commands can go after a command that is of the structure "sh aaa". Also abbreviations can be expanded to the full equivalent by using tab, so for example entering a command such as "acct-stop-cache" you can instead type in "acc" and press tab and the kernel will insert the full equivalent of "acct-stop-cache", if the word is ambiguous yu will probably need to add more characters to make it unique. If you make a typo and hit tab the kernel will print an error saying that the command expansion is invalid, and a marker ^ will indicate on which letter the command failed. If you do an incomplete command then an error will be printed saying that the command was incomplete.

2.5.5 Breaking out of commands with ctrl+c

If we type a command and we want to break out of it, by simply typing ctrl+c we can break out of that command and any process being executed by that command is now canceled.

2.5.6 More on the enable prompt and debug

The enable prompt is the prompt where we will most likely be working on. There are commands that allow you to debug information or processes. You can see all available debug commands by typing "debug ?" with a space - because we need a context sensitive help -. This will allow us to see information and see if they are running as intended. You cannot make changes to such information in the privilege exec prompt, to make changes you need to enter to another prompt called the global configurations prompt.

2.5.7 The global configurations prompt

To actually correct and change how processes and configurations are set we need to go to the global configuration which you can go there by typing "configure terminal" or its abbreviation "conf t". To see if you have entered into the other prompt correctly, the lines will now start with something like "Router(config)#"

2.6 Navigating the cisco IOS operating system Part 2

2.6.1 Command history

Lets enter a couple of random commands - you don't need to understand the commands - just to create some commands in order to be able to visualize the history.

```
1 -> Router(config)#ip host Server1 1.1.1.1
2 -> Router(config)#ip host Server2 2.2.2.2
3 -> Router(config)#Router1
                          ^
    % Invalid input detected at the '^' marker.
```

See the commands we typed in, the first one was ok, the second was ok, but the third one has an error. We need to fix the typo in command 3. We can surely type the command again and problem solved, but imagine it is a gigantic command and just one typo got an error and typing it again is going to be a waste of time. By simply hitting the up-arrow-key the last command will be re-written in the current line, there you can correct the typo and enter. If the typo is at the beggining of the command, to get to the beggining in one step you can type ctr+A which will put the mouse cursor in the begging of the line.

The command history saves all the commands you've typed in, it does not matter if they worked or not, they are saved in the command history. Therefore, by pressing the up-arrow-key you can put previous commands in the current line, and by pressing the down arrow key you can navigate downward through the history. The command history only shows the commands in the global configuration prompt, not on the enable prompt or in the user exec mode.

Remember the commands are valid depending on the command prompt level, you cannot do a command that is intended to be only runned in a certain prompt - such as the user exec prompt, etcetera - and then expect to use the same command in global configuration prompt or some other prompt. The command history comes in handy not just for typos but also if you make a mistake and type in a command at the wrong prompt level, therefore you can get to the right level and then by hitting the up-key you can run the same command and you don't waste time. For example:

```
Router(config)#hostname Router1
Router1(config)#show ip interface brief
                  ^
% Invalid input detected at the '^' marker.
```

We are at the wrong prompt layer, show commands need to have the proper privileges, we can give the command proper privileges by saying "do" at the beggining - remember you are not supposed to know what these commands do right now, it is just for show -. Press the up key and this will appear again:

```
Router1(config)#show ip interface brief
```

Put ctrl+a to go the the beggining of the line and type in "do".

```
Router1(config)#do show ip interface brief
```

Now it should work.

2.6.2 Prompts so far

So far we have seen that 3 prompts exists, and sort of seen how they are used and what they do.

1. User exec mode prompt
2. The priviliged exec mode - also known as the enable prompt -
3. The global configuration: used to set configurations for things that affect the router as a whole.

There are other levels, such as the interface configuration level which you can get to by typing:

```
Router1(config)#interfase fast fastEthernet 0/0
Router1(config if)#
```

2.6.3 Exiting pompts or 'dropping levels'

If you want to drop down a level you can say "exit" command, this will take you to the previous level. If you want to drop all the way back to the enable prompt you can simply type in the "end" command or by hitting the ctrl+c command.

2.6.4 Most common commands

- `show ip interface brief` : shows all the interfaces on the router, and the ip addresses as well as the protocol.
- `show running config` : shows the entire configuration on the router. Remember by hitting the space bar you can scroll through pages since this command does print out a lot of text.
- `sh run int fast0/0` : shows the configurations of a specified interface (int this case interface 0/0).
- `sh run | begin hostname`

  : this command use a 'pipe' (which we haven't seen), you can see if a command use a 'pipe' if the pipe symbol - which is '—' - is present in the command. In general, you can see the context of this command by simply putting a space and then a question mark after the pipe symbol, in this case the 'begin hostname' is simply to display the configurations running on that specific host name.

2.6.5 The IOS command line is not case sensitive except for one thing

Take into account that IOS is not case sensitive except for 'pipes' which use regular expression which is case sensitive. An example is:

```
Router1#sh run | begin Hostname
Router1#
```

After hitting enter to run the command the command printed no output, this is because 'Hostname' with a capital letter 'H' does not exit in the router so there is no output to show running configurations from. Pipes use REGULAR EXPRESSINS which you don't really need to know anything about yet except that they ARE CASE SENSITIVE. This is the only exception to the IOS command line. Pipes are the only exception to the IOS command line.

2.7 Cisco IOS Configuration management

In this section we will learn about IOS configuration management and running/startup configuration. It is important to realize that in IOS everytime you make a change the change takes effect immediately, the change goes into the running configuration which takes effect immediately, this is the running configurations which take in commands and go into effect immediately. Aside from the running configurations we also have the startup configurations - which are different from the running configurations - the startup configurations changes take effect when the router is restarted or rebooted, commands in the running config do not get saved permanently until you explicitly save them to the startup configuration, this is done so that you only make the changes take effect if you are sure they are correct. For example if you have a router and send configurations that are catastrophic but you have not saved it yet, you can simply restart the router and the changes that were catastrophic would simply never had taken place, this is a last resort.

2.7.1 Example of saving to startup configurations

Make sure you are in the user privilege exec prompt to do this, type "copy run start" and the kernel will ask you to enter a destination filename, it is common practice to stick with the default destination filename, and now if you hit "sh start" which means to show the startup configurations the changes were saved permanently.

2.7.2 Backing up configurations

*Sidenote: cisco does have an automated way of backing up configurations from a centralized server, but you can also backup manually as well. Simply type:

```
Router1#copy run ?
Router1#copy run flash:my-configuration
Destination filename [my-config]?
935 bytes copied in 1.236 secs (756 bytes/sec)
```

In this case we backed it up to a special location inside the router that is a flask drive location. If something were to happen we can retrieve our configurations from the configuration file backed up in the flash. To see if the configurations are indeed in the flash we can type the following command: "sh flash" (the name of the configuration file should be there). If we were to want to restore a back up we simple need to copy the back up to the start up config, but when you do a copy it is going to actually merge the commands rather than replace so typically you are going to want to replace the entire startup configuration, to do this you simply need to erase the startup configuration using the "erase start" - in newer routers and "wr erase" in older routers - once the startup config has been erased you can simply use the "copy flash:my-config start" to copy the backup to replace the startup configuration file.

2.7.3 Backing up outside the router

If the whole router blows up we would loose everything including the flash configurations, therefore it is a good idea to store those configs in a place where we can retrieve them even if the router blows up or anything of the sort. For this we can use a tftp server to backup the configurations. For this type the following commands:

```
Router1#copy run tftp
Address or name of remote host []? 10.10.10.10
Destination filename [router1-config]? router1-config-111111
```

The tftp server commands require that you have a tftp server to back up configurations, enter the IP address of the tftp server you have and then type in the name of the file you want, typically it is a good idea to save it as the routername followed by config followed by the date the backup is being performed.

2.7.4 How to check the contents of my backup configurations files

Use the the following command (replacing the configuration filename placeholder with your actual filename, in this case it would be "my-config"):

```
Router1#more flash:my-config
```

2.7.5 Configuration storage locations

- For the device to bootup it needs to load the IOS operating system image that is stored in flash.
  - The IOS operating system image is stored in Flash.
- Once the operating system is up and running you startup configurations will be loaded and processed.
  - The startup configuration is stored in NVRAM (non-volatile memory).
- The running configuration is stored in normal memory, which is RAM. It is loaded into RAM from the Startup Config when the device boots up.

Chapter 3

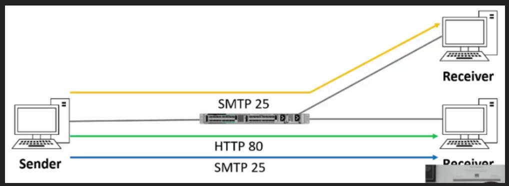
OSI Layer 4 - The transport layer

3.1 Introduction

- Port numbers, TCP, and UDP.

3.2 Layer 4 - The transport layer

- The transparent layer provides transparent transfer of data between hosts and is responsable for end-to-end error recovery and flow control.
  - Error recovery and flow control are not mandatory, some layer 4's will and others will not.
- Flow control is the process of adjusting the flow of data from the ender to ensure that the receiving host can handle all of it.
  - Example: If the sender is sending data at a faster rate than the reciever can handle then the flow control will have a mechanism that can go back to the sender to tell it to slow down.
- Session multiplexing is the process by which a host is able to support multiple sessions simultaneously and manage individual traffic streams over a single link.

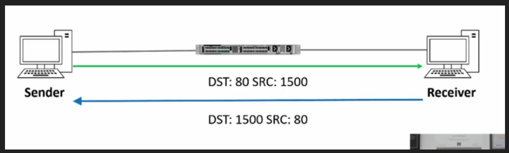


  - Notice that the sender has three sessions, has a SMTP session running in port 25, http session in port 80 and SMTP in port 25. It is the job of LAYER 4 THE TRANSPORT LAYER to keep track and control of all the sessions on our host.

3.2.1 Port numbers

- The layer 4 destination port number is used to identify the upper layer protocol.
- For example, HTTP uses port 80, SMTP email uses port 25, etc.
- The sender also adds a source port number to the layer 4 header.
- The combination of source and destination port number can be used to track session.

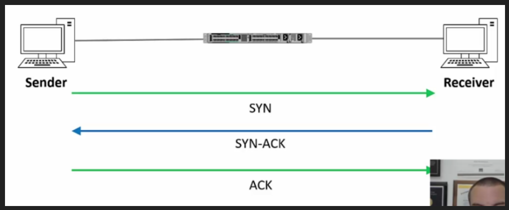
- For example, when a receiver host receiver a packet, it knows to which application to use it by its port number. If it comes in port 25 we know its comming for email application.



- When the sender sends the traffic to the receiver it will specify the destination and source port number, in the above example we are using port 80 and 1500, if traffic is expected to be returned then the receiver will flip the destination and source port numbers so that the source is now 80 and the destination is 1500.
- This is how state 4 firewalls are able to keep track of connections. We can restrict traffic in both directions or just in one, for example a firewall can restrict traffic from the receiver to the sender unless the sender has initiated communication with the receiver first.

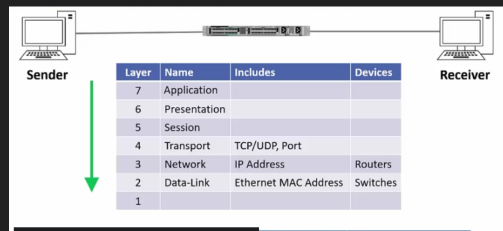
3.2.2 TCP

- TCP:
  - TCP is one of two most common protocols in layer 4. TCP (Transport Control Protocol) and UDP (the User Datagram Protocol) are the two most popular.
  - TCP is connection oriented: once a connection is established, data can be sent bidirectionally over that connection.
  - TCP is reliable: the receiving host sends acknowledgments back to the sender. Lost segments are resent. When a segment of data is lost the receiver will not send an acknowledgement to the sender and this will prompt the sender to resend the lost segment of data.
  - TCP performs flow control (remember its like synchronization between the sender and receiver so that the flow of data is received at a speed that is managable for the receiver).
  - TCP tends to have the opposite characteristics to UDP.
- The TCP Three-way handshake
  - The way that a TCP connection is set up is via the TCP three-way hand shake.

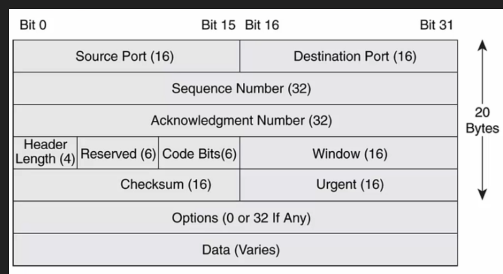


  - The way the three-way hand shake occurs is that it first sends a SYN message, SYN is short for synchronization message, when the receiver gets the SYN message it returns a SYN-ACK – which is shot for synchronization message acknowledgement – allowing it to complete the connection and now the two hosts have the connection and they can send traffic to each other, this is the 'third' step, that now that the sender has received a SYN-ACK they have established a connection and can send each other stuff.
- The TCP Header:
  -

- *Reminder: TCP packet header has all of the following:



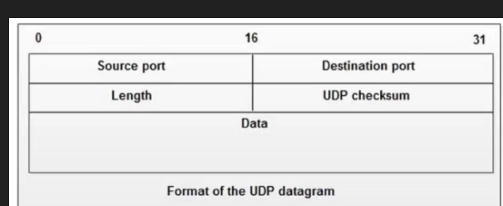
- Now a packet is assembled as follows:



  * We have all the fields represented by bits on a long word that comprices the packet. The first 16 bits of the packet stores the source port, the next 16 the destination port, and so on. This is very similar to how an assembler instruction is assembled in a microprocessor.

3.2.3 UDP

- UDP:
  - The UDP is User Datagram Protocol, it sends traffic best effort.
  - UDP is not connection oriented. There is no handshake connection setup between the hosts.
  - UDP does not carry out sequencing to ensure segments are processed in the correct order nor checks if they are missing.
  - UDP is not reliable: the receiving host does not send acknowledgements back to the sender.
  - UDP does not perform flow control.
  - If error detection and recovery is required it is up to the upper layers to provide it, it is NOT provided by UDP.
- The UDP Header:
  - In the UDP header we have much less fields compared to the TCP. Observe:



  - We only have the destination and source ports, length, UDP checksum, and the data.
  - This is both and advantage and a liability. With UDP you get much less overhead for example.

3.2.4 UDP vs TCP

- TCP and UDP have different use cases.
- Application developers will typically choose to use TCP for traffic which requires reliability.
- For example: real-time applications such as voice and video can't afford the extra overhead of TCP so they use UDP. If we were to use TCP on real time applications we would probably get large amounts of lagging and latency, because of the extra overhead, this would affect the quality and make it imposable to interact in real time, in such a case an unreliable connection with less header overhead is much appreciated.
- Some applications can use both TCP and UDP.
- TCP is the most commonly used protocol.

3.2.5 Common applications and their destination ports

TCP	UDP	TCP and UDP
• FTP (PORT 21) • SSH (PORT 22) • TELNET (PORT 23) • HTTP (PORT 80) • HTTPS (PORT 443)	• TFTP (PORT 69) • SNMP (PORT 161)	• DNS (PORT 53)

Chapter 4

OSI Layer 3 - The network layer

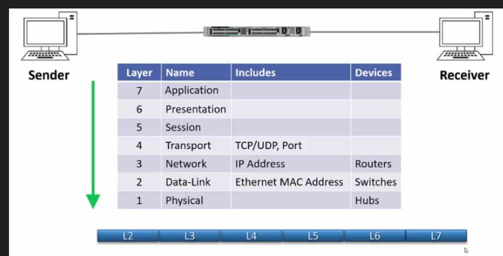
4.1 Introduction

- Format of the IP header.
- Format of IP addresses.
- Different types of traffic: unicast, broadcast, and multicast.
- Subnet mask.

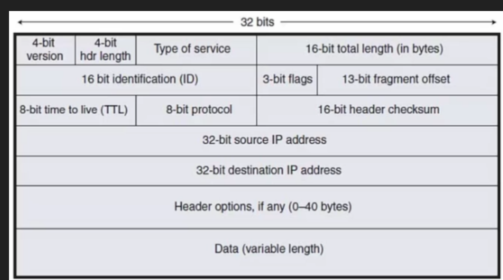
4.2 The IP header

- The network layer:
  - The network layer is responsible for routing packets to their destination and for Quality of Service.
    - Quality of service: when some levels of trafic require better quality of traffic, so for example, a voice or video is sensitive to delay so it will need a better quality service than something like email.
  - IP (Internet Protocol) is the best known Layer 3 Protocol. IPv4 is the focus of this section.
    - There is also IPv6 which is a newer protocol.
  - It is a connectionless protocol with no acknowledgements at Layer 3.
    - You can have reliable traffic by using TCP at layer 4, but itself has no acknowledgement. You can also have reliable traffic by building it into the upper layers.
  - Other Layer 3 Protocols include ICMP (Internet Control Message Protocol) and IPSec.
    - The ICMP is used for troubleshooting, IPSec is for secure and encrypted communications. There are others as well but these are the most common.
- IP addressing:
  - IP addressing is a logical addressing scheme which is implemented at layer 3.
  - The network designer uses IP addressing to partition the overall network into smaller 'subnets'.
  - This improves performance and security and makes troubleshooting easier.
    - It improves performance because instead of having a big network we have a set of small 'subnets' and we can keep the traffic to the particular subnet that it needs to be on rather than everywhere.
    - We also get better security. Example: if we have accounting servers, we can put it in it's own subnet and better control who has access to those servers.

    * Troubleshooting becomes easier because we now know that if a problem occurs it will show only on that particular subnet rather than in the entire network.
  - Layer 2 MAC addresses use one big flat addressing scheme. There is no logical separation between networks at layer 2, it's done at Layer 3.
    * IP addressing is a logical addressing scheme, whereas MAC addresses are a big flat scheme.
- OSI Reference Model - Encapsulation:



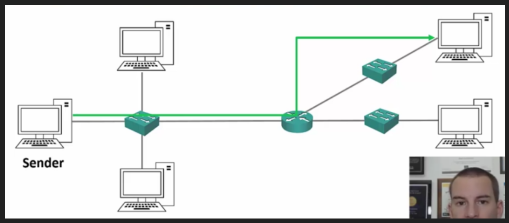
- The IP header:



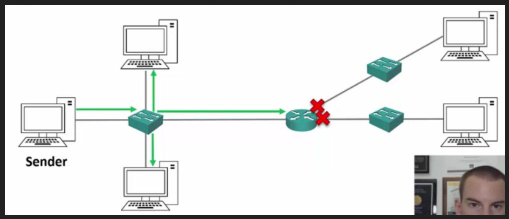
  - Version (4 bit): Ip version 4 or version 6.
  - header length: (4 bit): the length of the header.
  - Type of service: this is actually the quality of servie information, we can mark the packet with quality of service information so that we can make routers respond to this information.
  - total length (16 bit): length of the packet.
  - The second row of information (identification, flags, fragment offset) handles the fragments. Fragments: packets have a maximum size, when the information that is to be sent exceds the maximum packet size it is split up into several packets and sends them, these split up packets are called fragments.
  - time to live: everytime a packet goes through a router, the router will decrement the TTL by one, if it goes down to 0 the router drops the packet. This is to avoid routing loops, routing loops are when a packet is being sent through the same routers endlessly without being able to find its destination because of some error, to prevent this the time to live is decreased to be able to discard the packet once it has reached the maximum amount of times. This doesn't fix the problem in the network that caused the routing loop, but it will allow us to mitigate a potentially growing problem in our network.
  - 8-bit protocol: specify the layer 4 information type, usually TCP or UDP.
  - Check sum: to ensure that the packet has not been corrupted in transit.
  - source IP address (32 bit): from where the packet is comming from.
  - destination IP address (32 bit): to where it is headed.
  - Header options (0-40 bytes): this field is used to send any additional information in the header, it is not typically used.
  - Data (variable length): the actual data being transmitted.

4.3 Unicast, Broadcast, and Multicast Traffic

- Unicast, Broadcast, and Multicast Traffic
  - There are 3 main IP traffic types: unicast, broadcast, adn multicast.
  - Unicast traffic is to a single destination host.
  - Broadcat traffic is to all hosts on a subnet.
  - Multicast traffic is to multiple interested hosts.
- Unicast traffic:

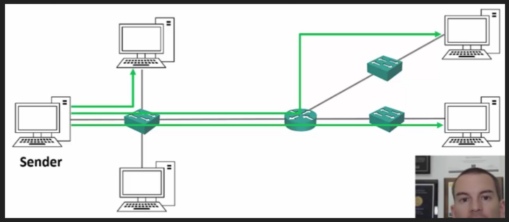


  - The traffic is coming from a sender, and is going just to one receiver.
- Broadcast traffic:

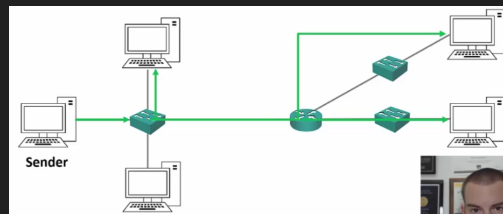


  - The sender sends ONE copy of the traffic that will come into the switch, and then it gets to all of the hosts connected to the switch.
  - *Important: when a broadcast is sent in a subnetwork, routers do not forward broadcast traffic, only switches do that.
- Unicast Traffic to Multiple Hosts:

  - With unicast you send out one individual copy of the requested data to the other hosts. This can potentially become expensive because of the extra bandwidth and can cause a lot of overhead.



  - When you unicast a message to several computers you send to each individual destination host a different copy. With multicast you send one copy that goes to several different receivers, with this arrangement you don't produce a different copy for all of the interested receivers, you send out one, and that one copy gets forwarded to all the different receivers.



– Multicast goes to multiple receivers but only if the receivers request it, with broadcast all hosts on a network get the data regardless of request, multicast is targeted.

– *Analogy: multicast is like tunning to a radio station, you have to tune in to get that particular radio station.

4.4 Converting from decimal to binary

- Counting in decimal:
  - Humans are conditioned to count in decimal.
  - For each 'column' in a number we have 10 possible choices, from 0-9.
  - Every time we add a column to the left, the value is multiplied by 10.
  - We start with '1s' as the further right column.

- 236 is two 100's, three 10's, and six 1's.

1000's	100's	10's	1's	
0 (236)	2 (36)	3 (6)	6 (0)	236

  - Writing 236 implies the following:

- Counting in binary:
  - Computers work in binary.
  - Electrical impulses are either off or on, so there's only two choices (0 or 1), unlike 10 in decimal (0-9).
  - For each 'column' in a number we have 2 possible choices, 0 or 1.
  - Everytime we add a column to the left, the value is multiplied by 2.
  - In this case the columns are like so:

- 236 in binary is 11101100

256	128	64	32	16	8	4	2	1
0 (236)	1 (108)	1 (44)	1 (12)	0 (12)	1 (4)	1 (0)	0 (0)	0 (0)
0	1	1	1	0	1	1	0	0

  - The number we are going to get in the end is 11101100. We do this like so. A final check is that at the end of the process you should have 0 left over.

4.5 IPv4 Addresses

- IPv4 Addresses:
  - An IPv4 address is 32 bits long.
  - It is written as 4 'octets' in dotted decimal format.
  - For example 192.168.10.15
  - Each octet is 8 bits long, 4 octets are 32 bits.
- Seeing your IP address:

  - If we were to type 'ipconfig' in cmd (windows) or 'ifconfig' in the terminal (linux and macos) we would get all sorts of information regarding the IP address.
  - IPv4 address is printed, the subnetmask is also printed, and the default gateway is printed. A default gateway is used when traffic is going to another host on another subnet, when the destination host is in the same subnet it can get there directly but if it is not it will go through a router and enter a different subnet, the default gateway in that case is the router to the other subnets and networks.
  - To see the default gateway on linux or macos enter 'ip route'.
  - To get your ip in Cisco OS, enter the enable prompt and type 'show ip interface brief'.
- Static and automatic addressing:
  - The IP address is usually se manually on servers, printers and network devices such as routers and switches. On desktop computers it is usually assigned automatically through the Dynamic Host Configuration Protocol (DHCP).
    - The reason you want to get your IP is set automatically on desktops is because you would not want to set it up manually because you would need to see which IP address is not in use and then set it up, instead you are automatically assigned one. With servers and routers you can do it manually because you they are fewer and they are going to almost always be the same IP address.
  - To understand how the logical separation between subnets works, you need to understand the IP address in binary.

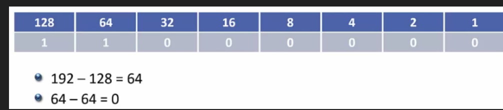
4.6 Calculating and IPv4 Address in binary

- Each octet in the IP address has a value ranging from 0 to 255.
  - Since an octet has 8 bits, and its binary we know that it has a maximum of 255.

128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

  - Notice if we replace the 8 bits with 0's we get 0. If we replace everything with 1 we get $128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$.
- Example: Let's convert that 192.168.10.15 address to binary, starting with the first octet of 192.

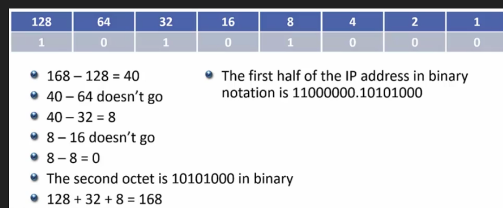
- Write out the binary columns on a piece of paper to do this: The final result is 11000000



128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0

- 192 – 128 = 64
- 64 – 64 = 0

- The second octet of 168.

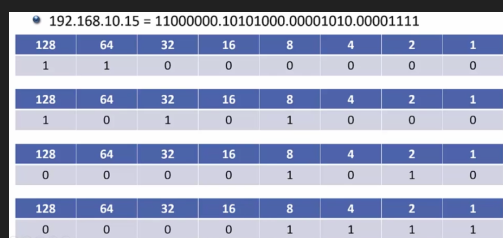


128	64	32	16	8	4	2	1
1	0	1	0	1	0	0	0

- 168 – 128 = 40
- 40 – 64 doesn't go
- 40 – 32 = 8
- 8 – 16 doesn't go
- 8 – 8 = 0
- The second octet is 10101000 in binary
- 128 + 32 + 8 = 168
- The first half of the IP address in binary notation is 11000000.10101000

- And so on.
- The final answer should be:

$$192.168.10.15 = 11000000.10101000.00001010.00001111$$



- 192.168.10.15 = 11000000.10101000.00001010.00001111

128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0

128	64	32	16	8	4	2	1
1	0	1	0	1	0	0	0

128	64	32	16	8	4	2	1
0	0	0	0	1	0	1	0

128	64	32	16	8	4	2	1
0	0	0	0	1	1	1	1

- To set the boundary between logical network (subnet), the IP address is combined with the subnet mask.
- The subnet masked has not been explained, it will be explained in the next section.

4.7 The subnet mask

- The subnet mask:
  - The subnet mask can be visualized by using the command prompt. You can enter the appropriate command to show you your IP and you will see your IP, default gateway, and it's subnet mask.
  - A host can send traffic directly to another host on the same subnet via switches.
  - For host to send traffic to another host in a different subnet, it must be forwarded by a router.
    * Routers are devices that link our different subnets together and route the traffic between them.
  - The host therefore sends to understand if the destination is on the same or a different subnet in order to know how to send it.
    * If the destination is on the same subnet it can send it there directly and no router is required. If the destination host is not in the same subnet then we must send it to a router, this router is the default gateway.
    * The way that the sender host knows if the destination host is on the same network is by simply comparing the sender's and reciever's IP address and subnet mask, if the mask is the same then just switches are required and no routing is done.

- The subnet mask is used for this.
- The subnet mask is also 32 bits long and can be written in dotted decimal or slash notation.

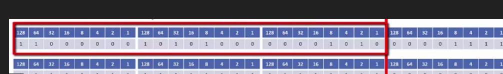
- Network and host portion:
  - A host's IP address is divided into a network portion and a host portion.
  - The subnet mask defines where the boundary is.
  - The easiest way to explain this is through example...

- Subnet 'Masking':
  - Let's say the host's IP address is 192.168.10.15 and its subnet mask is 255.255.255.0
  - We write the IP address out in binary notation, and then the subnet mask underneath.
  - Starting wit IP=192.168.10.15 and subnet mask=255.255.255.0 we convert to binary:

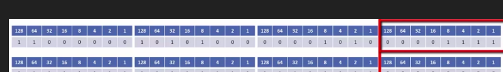
$$192.168.10.15 = 11000000\ 10101000\ 00001010\ 00001111$$
$$255.255.255.0 = 11111111\ 11111111\ 11111111\ 00000000$$

    * Notice the IP address, the IP address actuall contains part of the subnet mask, we just need to know where it starts and where it ends. The way we know when it starts is by simply seeint the subnet mask, whenever there is a 1 aligned in the position, this means that part of the IP is also the network address.
    * So in this example notice that the first 24 bits in the subnet mask are set to 1. This means that the first 24 bits – or from bit 0-23 – of the IP are part of the network address.
    * It is counter intuitive but just see how the numbers are linged up. We convert them to binary and align them up vertically as for the first bit of the IP address to correspond with the first bit of the subnet mask, same with the second, third, ... all the way up to the 32nd bit.
    * Now you see the subnet mask, wherever there is a 1, that is the network address.

    * In the picture above, notice the red line, the 1s come up to the third octet. To the left of that border lies the network address, to the right of that border lies the host address.
  - The IP address is compared ('masked') with the subnet mask.
  - A '1' in the subnet mask indicates that bit in the IP address is part of the network address.
  - A '0' indicates the bit is part of the host address.
  - Below you can see the network portion:



  - Below you can see the host portion:



- Local subnet or routed traffic:

– If the host wants to communicate with another host with an IP address which also begins with 192.168.10 (network portion for the example 192.168.10.20), it knows it's on the same subnet and it can send the traffic directly.

– If it wants to communicate with another host with any other network address (for example 192.168.11.20), it knows it has to send the traffic via a router.

– For a destination address to be in the same subnet, the network portion has to be exactly 192.168.10.

– Otherwise it's a in a different subnet and traffic must be sent via a router.

- Valid subnet masks:

  – The subnet mask always begins with contiguous '1's.

  – For example: 11111111.11110000.00000000.00000000 is a legal subnet mask.

  – 11101101.11110000.00001111 is not.

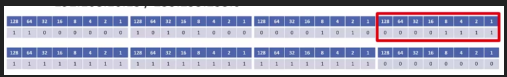
  – With IP addresses the '0's and '1's can be mixed, in the subnet mask it is always a '1's and then '0's, we don't mix them up in subnet masks.

- The host portion:

  – The host portion of the address is available to be allocated to the different hosts on the subnet (eg PCs, Servers, Printers, Router interfaces and Switch Management Addresses).

  – With two exceptions (that will be explained in a moment).

  – Again, this is the host portion of the address:



- Host addresses:

  – The host portion of the address specifies the individual host and must be unique on that subnet.

  – Hosts do not have to be numbered sequentially.

    * Any number can be assigned as long as they are not duplicated.

  – If the network portion of the address is 10.10.10.10, you can have a host with IP address 10.10.10.10 and another with 10.10.10.20.

  – You can't have two different hosts both with IP address 10.10.10.10. That would be a duplicate IP address. Whenever another host sent traffic to 10.10.10.10, the network wouldn't know which one to send it to.

  – We have host 10.10.10.10 on one subnet and host 10.10.20.10 on another subnet.

- The network address (network ID) FIRST EXCEPTION:

  – The first exception to the host portion is that all '0's in the host portion designates the network address and is not allowed to be allocated to a host.

    * Basically you can pick the host portion to be anything except all 0s because all 0s it is asumed to be the network address.

- In our example the network address is 192.168.10.0. Remember the network address is also 32 bits, it just so happens that when you have the host portion in all 0s, that has been reserved for the network address and you cannot allocate it to a host. In our example the network address is 192.168.10.0.

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0	1	0	1	0	1	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0
128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

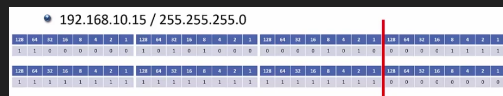
- The broadcast address SECOND EXCEPTION:
  - The second exception to host numbers are all 1s.
  - All '1's designates the directed broadcast address for the subnet.
  - Traffic with this destination address will be sent to all hosts in the subnet.
  - In our example the broadcast address is 192.168.10.255.
  - So all 0s and all 1s in the host's portion are the exceptions.

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0	1	0	1	0	1	0	0	0	0	0	0	0	1	0	1	0	1	1	1	1	1	1	1	1
128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

- Host addresses:
  - So taking out the exceptions of the host portion (all 1's and all 0's) that leaves the host portion free to be anything from 1-254. In our example that leaves 192.168.10.1 to 192.168.10.254 available to be allocated to hosts. All hosts connected to network address 192.168.10.0 can send traffic directly through switches because they are in the same subnet without going via their default gateway router.

4.8 Slash notation

- Subnet mask in slash notation:
  - So notice that the subnet mask is always contigous 1s followed by contigous 1s. This is long and tidious, there is a better way of writing it down. For IP address 192.168.10.15 and subnet mask 255.255.255.0 we can simply enumerate the 32 bits of the subnet mask and just write down the range from 1-n where the '1's are at.
  - Because the subnet mask always begines with contiguous '1's, it will be 1 to 32 bits ling counting from left to right.
  - This allows us to write the subnet mask in slash notation which is more convenient than dotted decimal for network diagrams or in conversation.
  - So look at the following example:

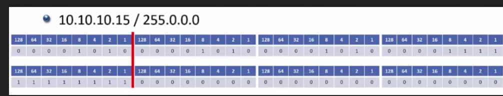


192.168.10.15 / 255.255.255.0

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0	1	0	1	0	1	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	1	1	1	1
128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

  - We see that the first three octets are 1s and the last octet is all 0s. So the slash notation for this case would be `/24` because the first 24 bits are set to 1, the remaining are 0s.
  - In cisco IOS the subnet mask is written in dotted decimal.

- Subnet mask in slash notation example 2:



- 10.10.10.15 / 255.0.0.0

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	1	1	1	1

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

  - This example can be written as either 10.10.10.15 255.0.0.0 or 10.10.10.15/8.
  - The network address is 10.0.0.0/8.
  - The last three octets is the host portion.
  - The available addresses would be 10.0.0.1 to 10.255.255.255.254, not 10.255.255.255 because that is the broadcast address.

Chapter 5

IP Address Classes

5.1 Introduction

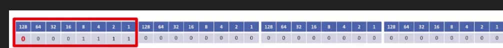
- Address classes A,B,C,D,E
- Important to understand for subnetting.

5.2 Class A IP Address

- Subnet Size:
  - Class A, B, and C IPs are used to allocate addresses to our hosts.
  - The bigger the host portion of the network, the more hosts we can have.
  - If the subnet mask is /8, we have 24 bits available to allocate to hosts.
  - If the subnet mask is /24, we only have 8 bits available to allocate to hosts.
- How internet addressing was meant to work:
  - The global coordination of internet IPv4 addressing is performed by IANA (Internet Assigned Numbers Authority).
    - Back when the designers were conceiving IPv4 they did not know that the internet was going to explode in popularity. There are some issues with IPv4 that would not make sense without context.
  - This is the way it was originally supposed to work:
  - When a company wants to communicate on the internet, they apply for a range of IP addresses.
  - If they have 6000 hosts, they ask for a range of IP addresses bug enough to convert that, plus room for growth.
  - They then allocate their addresses to their hosts in their various offices.
  - Unfortunately, when IPv4 was created, the designers didn't realise how big the internet was going to get, and they didn't create a big enough address space, this is why there is not enough addresses for everyone.
  - The long term solution to this problem is IPv6 which has a much bigger address space.
  - Private IP addresses with NAT (Network Address Translation) are currently deployed in the majority of enterprise networks as a workaround.
  - You'll learn all about private addresses, NAT adn IPv6 in a later lecture.

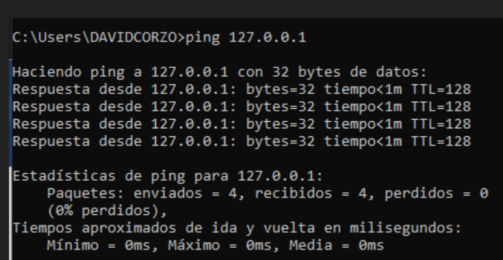
- To understand the lectures until we get to that point, think about it from the context of the originally intended IPv4 design, where all hosts which can communicate on the internet have public IP addresses.

- Class A:
  - The internet authorities split the IPv4 address space into separate classes.
  - Class A addresses are assigned to networks with a very large number of hosts.
    - In a class A there is going to be a small network portion and a large host portion.
  - The high-order (first) bit in class A address is always set to zero.
  - The default subnet mask is /8.
  - Valid networks addresses range from 1.0.0.0 to 126.0.0.0/8.
  - This allows for 126 networks and 16,777,214 hosts per network.

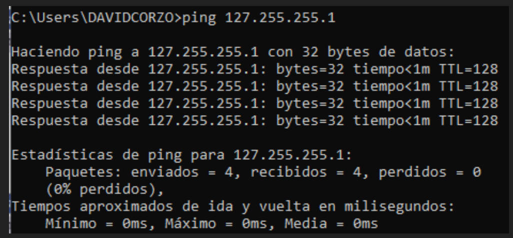


- Reserved Class A Addresses:
  - The same exceptions apply for host allocation, all zeros and all ones are reserved.
  - 0.0.0.0/8 is reserved and signifies 'this network'.
  - 0.0.0.1 to 0.255.255.255 are not valid host addresses. Because the network portion is all 0s, as long as the network portion is 0 the rest cannot be used.
  - 127.0.0.0/8 in the Class A space is reserved as the loopback address for testing the local computer.
  - 127.0.0.0 to 127.255.255.255 are nto valid host addresses.
    - 127.0.0.0 is used to test the IP stack on the local computer.
  - This means that because the first 8 bits cant be used when they are all set to 0 or when they are all set to 1, any combination of the host part is invalid if it begins in all 0s. This wiped out 33,554,428 addresses from the global address pool – whops!
  - They could have used just two address for this, but instead they chose the convention of all 0s and all 1s and ended up wiping out 16 million addresses twice. This is because the designers did not know that the internet was going to explode.

- Using the loopback:
  - Type ping 127.0.0.1 and you will ping the local IP stack:



  - But since the entire network address starting in 127 is reserved any combination of digits after the 127 are going to be ignored and you would be pining the local IP stack, this is why we lost 16 million IP addresses. So here I can invent any number, as long as it starts with 127.X.X.X it will reference the same reserved loopback address.



```
C:\Users\DAVIDCORZO>ping 127.255.255.1

Haciendo ping a 127.255.255.1 con 32 bytes de datos:
Respuesta desde 127.255.255.1: bytes=32 tiempo<1m TTL=128
Respuesta desde 127.255.255.1: bytes=32 tiempo<1m TTL=128
Respuesta desde 127.255.255.1: bytes=32 tiempo<1m TTL=128
Respuesta desde 127.255.255.1: bytes=32 tiempo<1m TTL=128

Estadísticas de ping para 127.255.255.1:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 0ms, Máximo = 0ms, Media = 0ms
```

- Subnetting:
  - Obviously a company wouldn't put all 16,777,214 hosts into a single logical network, this would be terrible for performance and security.
  - They would split their /8 address allocation into smaller subnets and allocate these to different offices adn types of hosts.
  - For example if they recieved 15.0.0.0/8, they could allocate the subnet 15.0.1.0/24 to sales computers in New York, 15.0.2.0/24 to accounting PC's and 15.0.9.0/24 to sales computers in Boston.
  - This is called subnetting and you'll master it later in this section.
  - So they would have that huge network that they got assigned by the internet authorities and they can now assign it to their different offices and hosts. This is called subnetting.

5.3 IP addresses class B and C

- Class B:
  - Class B addresses are assigned to medioum-sized to large sized networks.
  - The two high-order bits in a class B address are always set to binary 1 0.
  - The default subnet mask is /16.
  - Valid network addresses range from 128.0.0.0 to 191.255.0.0/16.
  - This allows fro 16,384 networks and 65,534 hosts per network.
  - This would also be subnetted in a real world environment.

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

  - 131.192.0.0/16.

- Class C:
  - Class C addresses are used for small networks.
  - The three high-order bits in a class C address are always set to binary 1 1 0.
  - The default subnet mask is /24.
  - Valid network addresses range from 192.0.0.0 to 223.255.255.0/24.
  - This allows for 2,097,152 networks and 254 hosts per network.
  - This could be allocated as is for a real world network, or subnetted into smaller subnets.

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

    * 195.0.192.0/24

- A quick note on private addresses:

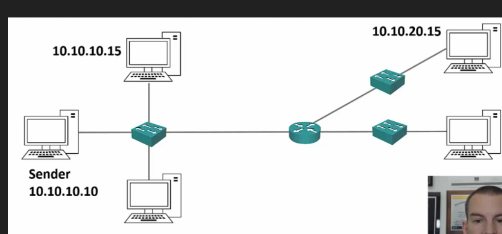
- There is also a range of reserved private addresses in each class.
- These are valid to be assigned to hosts but they are not routable on the public internet.
- They were originally designed for hosts in a closed private network with no internet connectivity.
  * For example in a highschool that has a computers class that does not want them to browse the internet, rather just communcate with each other, these would be put on a private address.
  * Another perk is that private IP addresses are free, a public IP you need to pay.
- Ranges used for private addresses are:
  * Class A: 10.0.0.0 to 10.255.255.255
  * Class B: 172.16.0.0 to 172.31.255.255
  * Class C: 192.168.0.0 to 192.168.255.255
- Private addresses will be discussed in a later lecture in this section.

5.4 IP address classes D and E

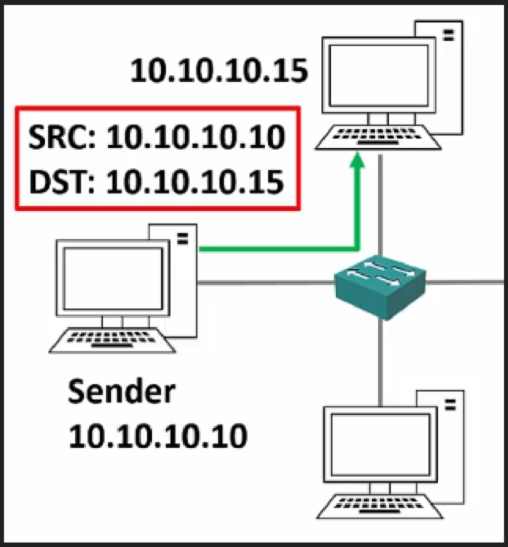
- Address classes:
  - Class A, B and C include all the addresses with are valid to be assigned to hosts.
  - What about 244.0.0.0 to 255.255.255.255?
- Class D:
  - Class D addresses are reserved for IP multicast addresses.
  - To four high-order bits in a class D address are always set to binary 1 1 1 0.
  - These addresses are not allocated to hosts and there is no default subnet mask.
  - Valid addresses range from 224.0.0.0 to 239.255.255.255

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	0	0	0	1	1	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	1

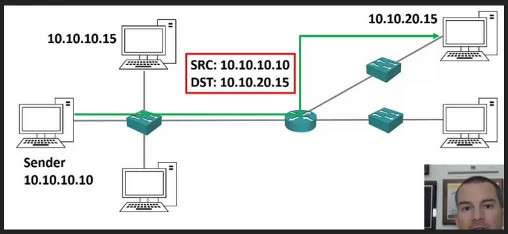
  - 227.1.192.5
- Reminder on unicast traffic:



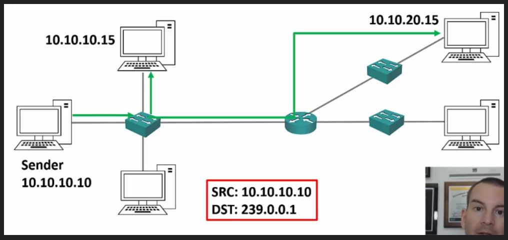
  - On unicast traffic the sender (10.10.10.10/24) sends something via the network to one of the receivers on the network 10.10.10.15/24. The information will travel independently to him using the layer 3 ip and since they are in the same subnet, they can communicate directly. So we unicast the traffic to the destination host via a switch.



- Now lets suppose the source host wants to send exactly the same traffic to another host on another network 10.10.20.15/24, the traffic will now be unicast via a router to the destination host. Notice that the traffic is being sent independently to each host even though its entirely the same traffic, if we had 1MB traffic unicast to each device we would use 1MB for exactly the same material, what we want is to send it once and the same traffic to be reused to the next host accordingly.



- We can improve this by simply using multicast traffic instead of unicast. With unicast we send the traffic to a reserved address, and from there it forwards it to the destination hosts, and it only gets sent once, it does not consume as much bandwidth because the traffic is just sent once.
- So here is how this works: in the sender a special application is going to send the traffic to a special and reserved destination called a multicast address, that in our network will be 239.0.0.1, it comes from the same source address, and the destination addresses interested in the traffic will get the traffic, but the difference lies in that this time we only send it once instead of various times. This is better.



- An analogy Neil describes is like tunning into a radio station, the radio station is a multicast because all interested destination hosts tune into it, the frequency they tune into is like the special multicast address we described earlier. By tunning in to there, they will get the traffic that is forwarded.

  - Here if there were 50 interested hosts the bandwidth required would still be just 1MB (assuming the traffic was 1MB) it would not be 50MB like we would get with unicast, but it would be only 1MB. Much more efficient and cheaper.

- Class E:
  - This class E addresses are 'experimental and reserved for future use'.
  - The high-order bits in a class E addrfess are set to 1111.
  - These addresses are not allocated to host and there is no default subnet mask.
  - Addresses range from 240.0.0.0 to 255.255.255.255.
  - 255.255.255.255 is the broadcast address for 'this network'.

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	0	0	1	1	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	0	1	0

  - 243.0.192.10
  - Class E addresses are really just experimental, class D is used in multicast, but you need to know both especially for the CCNA.

- IP address class summary:

Class	First octet	Default subnet mask
A	1-126	/8
B	128-191	/16
C	192-223	/24
D	224-239	
E	240-255	

  - Class A,B, and C can be assigned to hosts.
  - Class D is for multicast.
  - Class E is experimental.

Chapter 6

Subnetting

6.1 Introduction

- Last section was about OSI layer 2, IP addresses, subnetmasks, etc.
- In this section we are still in layer 3 and we are going to go much deeper, and learn to control the boundaries.

6.2 CIDR - Classless Inter-Domain Routing

- CIDR - Classless Inter-domain routing:
  - A problem with classful addresses was that if a company had more than 254 hosts they would need to be assigned a class B network.
    - The internet authorities gave class A,B,C,D,E addresses. If they had 500 hosts they would get addresses for 65,543 hosts because the next class would allocate the whole next octet of the host portion. This was a waste.
  - They would have much less than the 65,434 hosts allocated, so this wasted a huge amount of the global address space.
  - Classless inter-domain routing (CIDR) was introduced in 1993 to alleviate this problem.
    - This removed the problem of classful addresses only allocating octets and instead allowed them to be split into smaller networks.
  - CIDR removed the fixed /8, /16, and /24 requirements for the address classes, and removed them to be split or 'subnetted' into smaller networks. For example 172.10.10.0 /20.
  - Companies can now be allocated an address range which more closely matches their needs and does not waste addresses.
  - The name for this is **subnetting**.
  - The benefit of Subnetting is that we don't waste addresses.
- CIDR and Route summarisation:
  - Another benefit of CIDR is that aggregate blocks of networks can be advertised on the internet.
  - For example lets say that a company called 'ISP A' gets assigned from the internet authorities the range of IP addresses ranging from 175.10.0.0/24 - 175.10.255.0. We also have another company called 'ISP B' are given 175.11.0.0/24 - 175.11.255.0/24.
  - Now these companies connect to each other and when CIDR comes in that ISP A and ISP B would need to copy all the routes from its routers and advertise all their 256 address blocks to each other, but when we have CIDR we just summarize the bits that they have in common so that we dont have duplicate data in the routing tables.

– In this case ISP A would advertise 175.10.0.0/16 to ISP B, and now ISP B only learns one router to get to ISP A rather than learning 256 routes. This is complicated and vaguely explained in this lecture, so watch the youtube video below.

– This was actually not very well explained in the course so this youtube video actually cleared it up: `https://www.youtube.com/watch?v=YOpwKwol8xk`

- Route summarization benefits:
  - ISP A does not know about all 256 /24 networks reachable in ISP B.
  - It only has the single 175.11.0.0/16 summary route. It has 1 route instead of 256. Less memory is used.
  - This reduces the size of ISP A's routing table and takes up less memory.
  - If an individual link goes down in ISP B, it has no impact on ISP A. The single summary route does not change.
  - (Routers in ISP B would have to recalculate their routing table if a link went down)
  - This restricts issues to the local part of the network and reduces CPU overload.

6.3 Subnetting overview

- Subnetting:
  - Subnetting is one of the core knowledge areas in networking, crucial for the CCNA and for networking in general.
  - To understand this lecture, think about it from the point of view of the originally intended IP4 design again, where all hosts which can communicate on the internet have a public IP address.
    - ∗ Think of it 10-15 years ago when people did not have problems with IP address shortage.
  - Let's say we're the network designer fro a small business with four departments spread over two offices, and we want to manage our own public address space.
    - ∗ We could buy 4 separate class C networks for that. But the problem is that public IP addresses are expensive.
  - Rather than purchasing separate address ranges for the different departments, we can purchase a single class C range and subnet it into smaller portions. And then we can subnet that to the different departments of our network.
- Borrowing host bits:
  - Let's say we've been allocated class C 200.15.10.0/24.

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	0	0	1	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

  - To subnet the network into smaller subnets, we need to 'borrow' host bits and add them to the network portion of the address.
    - ∗ We are going to move the border of the subnetmask to the right.
    - ∗ WHEN WE ARE DOING SUBNETTING THE LINE ALWAYS MOVES TO THE RIGHT.
  - The network address line always moves to the right when we subnet.
  - The further to the right we go, the more subnets we'ññ have of that size but less hosts.
- To calculate the number of networks:
  - To calculate the number of available subnets, the formula is $2^{\text{subnet-bits}}$.

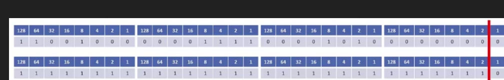
  - If a class C network uses a 28 subnet mask then we've borrowed 4 bits from the default of /24.
  - $2^4 = 16$ available subnets.
  - If a class B network uses a /28 subnet mask then we've borrowed 12 bits from the default of /16.
  - $2^{12} = 4096$ available subnets.
  - Hosts on different subnets need to go via a router if they want to communicate with each other.

- To calculate the number of hosts:
  - To calculate the nuber of available hosts, the formula is $2^{\text{host-bits}} - 2$.
  - We subtract 2 because the network address and broadcast address cannot be assigned to hosts.
  - If a class C network uses a /28 subnet mask then we have 4 bits left for hosts.
  - $2^4 - 2 = 14$
  - If a class B network uses a /28 subnet mask then we have 4 bits left for hosts.
  - The amount of hosts we get is going to be dependant on the subnet size.

- A quick note on `ip subnet-zero` command:
  - Just like we have to subtract 2 to get the number of valid hosts, we used to have to subtract 2 to get the number of available networks also.
  - In the original internet standards, it was not allowed to use network bits of all 0's or all 1's (just like we can't use all host bits of all 0's or all 1's).
  - There wasn't really any practical need for this and it wasted address space.
  - The `ip subnet-zero` command on a router overrides the limitation, and is enabled by default.
  - This is important because on the updated standards we don't use the 2 subnets we use to take off for the network, we ONLY DO THIS FOR THE HOSTS. Hence, you DO NOT SUBTRACT 4, you subtract 2. Do not do that for the CCNA.

6.4 Subnetting classs C networks and VLSM

- Class C /31 subnet:
  - Let's say we've been allocated class C 200.15.10.0/24.



    * If we want to have any hosts available then the highest subnetmask we can have is /31. In that case we leave 1 bit remaining for the hosts.
  - If we move the line all the way to the right we're now using /31 (or 255.255.255.254).
  - This leaves one bit for the host address, with a possible value of 0 or 1.
  - It borrows 7 bits from the network address.
  - This gives us 128 subnets 2^7 which accommodates 2 hosts each.
  - We subnet using /31. Valid host addresse:
    * 200.15.10.0 to 200.15.10.1
    * 200.15.10.2 to 200.15.10.3
    * etc. to:
    * 200.15.10.254 to 200.15.10.255
  - Notice that the first three octets never change. The subnet mask changes in the first 7 bits.

– What about the network and broadcast address?
  * /31 breaks the standard rules of IP addressing.
  * /31 subnets are supported in cisco routers for point to point links (which have no need for a network or broadcast address).
    · A point to point link means that all traffic moves from one to the other in terms of all messages sent can only be sent to the one other host in that subnet. There is no other place for the message to go so this is why we don't need a broadcast and network address.
    · This is why cisco allowed that exception, we can have a /31 subnet mask on a point to point link.

- Class C /31 subnet mask:
  – Let's say we've been allocated a class C 200.15.10.0/24.
  – Let's move the line back a place. We're now using /30 (or 255.255.255.252).
  – This leaves 2 bits for the host address, $2^2 - 2 = 2$ hosts available and the network and broadcast address.
  – It borrows 6 bits for the network address.
  – This gives us 64 subnets 2^6 which accommodate 2 hosts each.
  – Notice that the line is after the 4. The network addres goes up in values of 4.

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	0	0	1	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0
128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0

  – Valid hosts addresses:
    * 200.15.10.1 to 200.15.10.2 (network .0, broadcast .3).
    * 200.15.10.5 to 200.15.10.6 (network .4, broadcast .7).
    * etc. to:
    * 200.15.10.253 to 200.15.10.254 (network .252, broadcast .255).
  –

6.4.1 /31 vs /30

- /31 and /30 both accommodate 2 hosts per subnet. Because /31 subnets dont have a network and broadcast address.
- /31 supports 128 subnets, /30 only 64.
- /31 us useful if you need to maximize use of your address space.
- /30 is more standard and commonly used.
- For the CCNA exam, use /30 when a subnet to support 2 hosts is required, unless told to use /31.
  – Because usually a /30 is the correct answer. Only to maximize use /31, otherwise always use /30.

6.4.2 Class C /29 subnet

- Let's say we've been allocated Class C 200.15.10.0/24.

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	0	0	1	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0
128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0

- Let's move the line back a place. We've now using /29 (or 255.255.255.248).
- This leaves 3 bits for the host addresses, $2^3 - 2 = 6$ possible hosts.
- It borrows 5 bits for the network address.
- This gives us 32 subnets 2^5 which accommodate 6 hosts each.
- The valid host addresses are:
  - 200.15.10.1 to 200.15.10.6 (network .0, broadcast .7)
  - 200.15.10.9 to 200.15.10.14 (network .8, broadcast .15)
  - etc. to:
  - 200.15.10.249 to 200.15.10.254 (network .248, broadcast .255)

6.4.3 Other class C subnet masks

- We can carry on moving the red line back a place.
- We can do for example:
  - /28 or (255.255.255.240) = 16 networks of 14 hosts each.
  - /27 or (255.255.255.224) = 8 networks of 30 hosts each.
  - /26 or (255.255.255.192) = 4 networks of 62 hosts each.
  - /25 or (255.255.255.128) = 2 networks of 126 hosts each.
  - /24 or (255.255.255.0) = 1 network of 254 hosts (the default, no subnetting).

6.5 Subnetting practice questions

- What are the network address, broadcast address, and valid host address for the IP address 198.22.45.173/26?
  - 198.22.45.173 = 11000110.00010110.00101101.10101101
  - The highlighted part is the /26, we cannot do anything with those bits, only with the reamining.
  - The network and broadcast addresses:

Network address	Broadcast address
11000110.00010110.00101101.10000000	11000110.00010110.00101101.101
198.22.45.128	198.22.45.191

  - The valid host addresses: 198.22.45.129 - 198.22.45.190.
- What is the subnet mask in dotted decimal notation?
  - 255.255.255.192
- Note that when we subnet a class C address the magic is all going to happen in the last subnet.
- So we didn't really need to write out the 198.22.45 part.

6.6 Variable Length Subnet Masksking Example 1

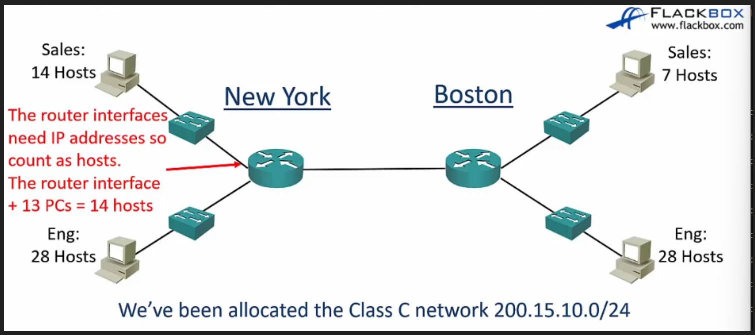
- Early routing protocols only supported fixed length subnet masking (FLSM) where all the subnets had to be the same size. You couldn't have a subnet with 14 hosts and another with 64 hosts in the same network.
  - If you had a subnet that was /28 they all had to be /28 no /27 or any other submask within the same larger network.
  - Now variable length subnet masks are supported.
- All modern routing protocols support variable length subnet masking. This allows us to size subnets differently according to how many hosts they have.

6.6.1 Subnetting considerations

- How many locations do we have in the network?
- How many hosts are in each location?
- What are the IP addressing requirements fro each location? (Should different department types of host be in different subnets?)
- What size is appropriate for each subnet? (Don't waste addresses, but leave room for growth)
  - Don't make it too big, and don't make it too slow as to have to make a topology redesign later.

6.6.2 Network topology diagram

- We have a network topology diagram, we have an organization with an office in NY, another one in Boston, NY is the headquarters, they have 14 hosts in the sales department and 28 hosts in the engineering department. In boston they have 28 in engineering and 7 in the sales department.



- We've been allocated the class C network 200.15.10.0/24 from our internet service provider.
- We have Point to point link between the routers, this means that each of them are going to get one of the host ip addresses.

6.6.3 Subnetting design steps

- Find the largest segments and allocate a suitable subnet size for it.
- Allocate this subnet at the start of the address space.
- Continue going down the list.
- In the real world you want a scalable design – you will likely allocate spare subnets for future growth, and leave space in the subnets for additional hosts.
  - Example: the engineering department we have 28 hosts, don't give the subnet that size because you never know if you might need mor in the future.
  - Maybe you need 2 more hosts in the next month for new employees, you need to keep this in mind.
  - And if you allocate 2 subnets of 30 for the engineering and sales, don't do this, instead leave 3 or 4 subnets of 30 hosts, because that way you have the capacity for growth and you don't need to redesign the topology later on, everything is still sequential and contigous and everything works.
  - Don't design for what is required right now, leave some room for growth.
  - THIS IS WHAT YOU DO IN THE REAL WORLD, IN THE EXAM DON'T THINK LIKE THIS. DO EXACTLY AS YOU ARE TOLD. Even if you think that would be a stupid thing to do.

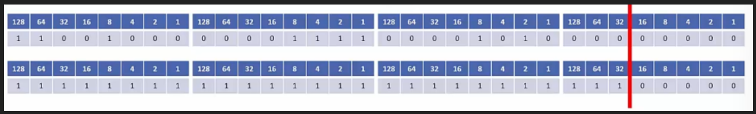
- In the CCNA exam do exactlu what the question asks, don't worry about wether it's best practice or not.

6.6.4 Engineering departments

- The engineering deparments in both sites have 28 hosts.
- For our example we've been told that the departments will not grow and we need to use the smallest subnets possible to maximise our address space.
- Let's find the optimal subnetmask for the engineering departments, and determine the network and broadcast addresses of each.

6.6.5 Engineering departments answer

- We've been allocated 200.15.10.0/24


128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	0	0	1	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0
128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0

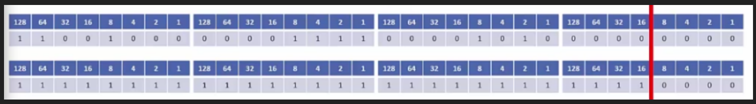
- /27 or (255.255.255.224) supports 30 hosts. It is not 28 because we need the network address and the broadcast + the 28 hosts.
- New york engineering subnet:
  - Network address: 200.15.10.0/27
  - Broadcast address: 200.15.10.31
  - Hosts 200.15.10.1-30.
- Boston engineering subnet:
  - Network address: 200.15.10.32/27
  - Broadcast address: 200.15.10.63
  - Hosts: 200.15.10.33-62

6.7 Variable Length Subnet Masking Example Part 2

- We will go through the subnet for the rest of the departments.

6.7.1 New York Sales Department

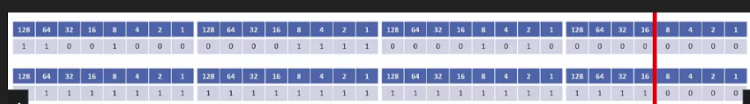
- The question:
  - The next largest subnet is New York Sales which requires 14 hosts.
  - Pause the video here and calculate the optimal subnet mask.
  - Also determine the network and broadcast addresses that we will allocate, and the range of host addreses.
- We've allocated 200.15.10.0/24


128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	0	0	1	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0
128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0

- /28 (0r 255.255.255.240) supports 14 hosts.
- 200.15.10.0 to 200.15.10.63 are already in use by the engineering departments, so this network address will start at 200.15.10.64.
- The network addresses goes up in values if 16, so the next one is 200.15.10.80.
- Our broadcast address is 200.15.10.79.
- Valid hosts addresses are 200.15.10.65 to 200.15.10.78.

6.7.2 Boston Sales Department

- The question:
  - The next largest subnet is Boston Sales which requires 7 hosts.
  - Pause the video here and calculate the optimal subnet mask.
  - Also determine the network and broadcast addresses that we will allocate, and the range of host addresses.
- We've allocated 200.15.10.0/24



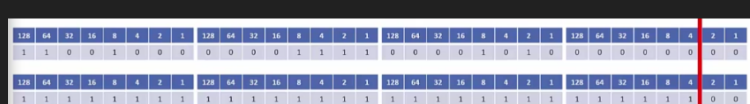
- /28 supports 14 hosts.
- /29 does not support 8 hosts because of the network and broadcast address.
- 200.15.10.0 to 299.15.10.79 are already in use, so this network address will start at 200.15.10.80.
- The network address goes up in valies of 16, so the next one is 200.15.10.96.
- Our broadcast address is 200.15.10.95.
- Valid host addresses are 00.15.10.81 to 200.15.10.94.

6.7.3 Subnetting

- We are not done.
- We need to link the routers. Also need to allocate address space for the router loopback interface (not yet seen in the course).

6.7.4 Point to point links

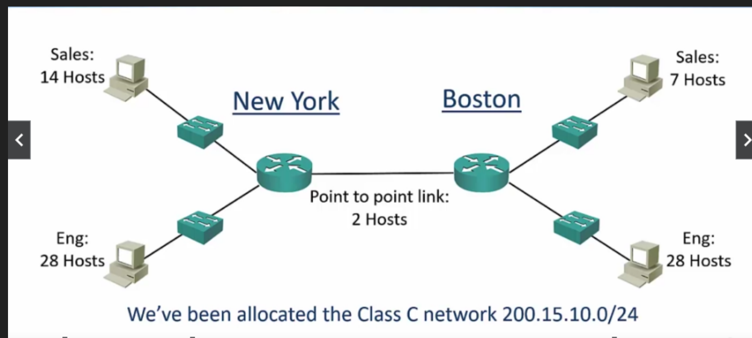
- We've allocated 200.15.10.0/24



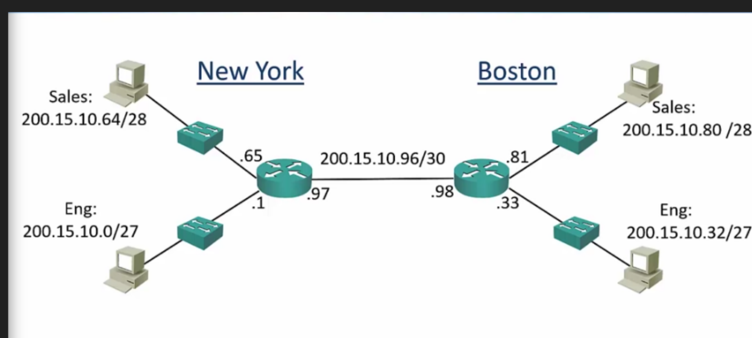
- /30 supports 2 hosts.
- 200.15.10.0 to 200.15.10.95 are already in use, so out network addresses will be 200.15.10.96.
- The network address goes up in values of 4, so the next one is 200.15.10.100.
- Our broadcast address is 200.15.10.99.
- Valid host addresses are 200.15.10.97 to 200.15.10.98.

6.7.5 Network topology diagram

- Revisit the network diagram again:



- Update the diagram with the IP address information for each subnet.
- Also enter the IP address on the router.
- The default gateway address should be the first available address in each subnet.
- It looks like this:



6.8 Subnetting Large Networks Part 1

- Given the class B 135.15.0.0/16

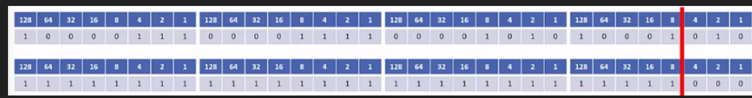
128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	0	0	0	0	1	1	1	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

- If we subnet this in /29 subnets, we have 3 bits for host addressing. This allows 6 hosts per network ($2^3 - 2$), the same as if we used /29 with a class C address.
- Because we were allocated a Class B /16 address range, we have 13 bits for network addresses. This allows for 8292 subnets (2^{13})

6.8.1 Examples

- Example 1: For the IP address 135.15.10.138/29, what is the network address, broadcast address, and range of valid IP addresses.

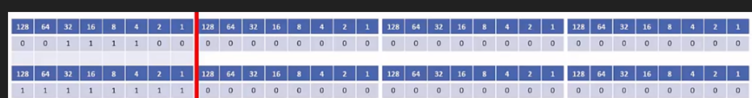


- Notice in the above figure, the lower binary word is the subnetmast, since it has a contigous stream of 1's and then three zeros. The upper is the ip address.
- Thus the broadcast address is simply the ip in which the last 3 binary digits is 1, the network is when the last three digits are 0:
    - The base IP: 135.15.10.138 = 10000111.00001111.00001010.10001010
    - Network address: 10000111.00001111.00001010.10001000 = 135.15.10.136
    - Broadcast address: 10000111.00001111.00001010.10001111 = 135.15.10.143
    - Valid hosts: from the network address to the broadcast address - 2: 10000111.00001111.00001010.10001001 - 10000111.00001111.00001010.10001110 = 135.15.10.137 - 135.15.10.142

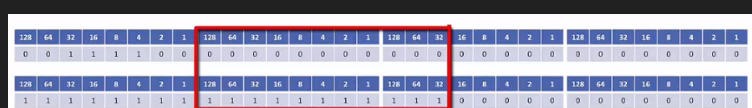
6.9 Subnetting Large Networks

Example: class A on the third octet.

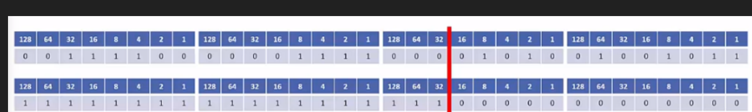
- We are allocated a class A 60.0.0.0/8:



    - If we subnet this into fixed size subnet masks using /19, how many subnets do we have?
        - The answer is /19 means we have 32-19 bits, or 13 bits for host addressing. This means we can reference up to $(2^{13} - 2)$.
        - This because we were allocated a class A /8 address range, we have 11 bits for network addresses, or $2^{11} = 2048$ subnets.



- The IP address 60.15.10.75/19, what is the network address, broadcast address, adn range of valid IP addresses?



    - Notice that the line is after the 32, so the network address goes up in multiples of 32.
    - Network address: 60.15.0.0
    - Next network address: 60.15.32.0
    - Broadcast address: 60.15.31.255
    - Valid host addresses: 60.15.0.1 - 60.15.31.254.

- You have been asked to subnet the 134.65.0.0 network into six different networks.

	FIRST OCTET									SECOND OCTET									THIRD OCTET									FOURTH OCTET							
subnet index	1	2	3	4	5	6	7	8	.	9	10	11	12	13	14	15	16	.	17	18	19	20	21	22	23	24	.	25	26	27	28	29	30	31	32
ip addr dec	134								.	65								.	0								.	0							
ip addr bin	1	0	0	0	0	1	1	0	.	0	1	0	0	0	0	0	1	.	0	0	0	0	0	0	0	0	.	0	0	0	0	0	0	0	0
subnet mask de	255								.	255								.	255								.	248							
subnet mask bi	1	1	1	1	1	1	1	1	.	1	1	1	1	1	1	1	1	.	1	.	1	1	1	1	1	1	.	1	1	1	1	1	0	0	0

- 134.65.0.0 is a class B network so we'll be subnetting on the third octet to get 6 subnets.
- 3 bits would provide 8 subnets 2^3
- So the answer is /19 or 255.255.224. Since the subnet mask provided by class B ip addresses specify that the network address is the first 2 octets, we extend the subnet mask by three bits of the third octet the required 8 subnets.
- Network addresses would be 134.65.0.0, 134.65.32.0, 134.65.64.0, etc.
- 8190 hosts in each subnet $2^{13} - 2$

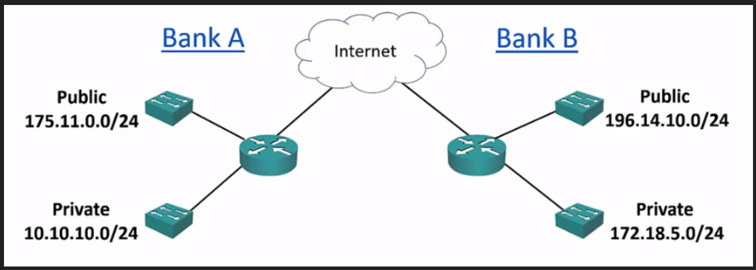
6.9.1 Subnetting question categories

- Most exam questions will have the following format:
  - Given a network requirement of 'x' amount of subnets and 'y' amount of hosts per subnet, what network addresses and subnet mask should be used for each subnet?
  - Given a particular IP address and subnet mask, calculate:
    - The subnet's network address.
    - The broadcast address.
    - The range of valid host IP addresses.

6.10 Private IP addresses part 1

- Private addresses were documented by the IETF (Internet Engineering Task Force), when this organization documents anything it issues a RFC which stands for Request For Comment, basically its calling for peer reviewing of the documentation, so other engineers can comment and suggest improvements.
- The RFC 1918 specifies private IP address ranges which are not routable on the public internet.
  - It is not common to memorize RFC numbers, but most engineers just know about RFC 1918 for private addresses, it is common terminology.
- RFC 1918 Private Addresses:
  - When the internet was thought out, there was some situations in which you wanted some IP addresses to not be connected to the internet.
  - For example in a company, you have a LAN that is not connected to the internet.
  - So private addresses were designed to have no connectivity to the internet.
  - A publich IP, in contrast, costs money. So if you don't require IPs to connect to the internt it would be a waste of money and a potential vulnerability threat to use public. Instead use private.
  - An example of organizations that use this are banks: it makes sense that banks would want to a part of their network to be completely cut off the internt for security reasons. They would use private IP addresses there.
  - There is a range of private addresses in each address class:
    - 10.0.0.0 - 10.255.255.255: class A /8 subnet mask.
    - 172.16.0.0 - 172.31.255.255: class B /12 subnet mask.
    - 192.168.0.0 - 192.168.255.255: class C /16 subnet mask.
- Example of when we would use private addresses:

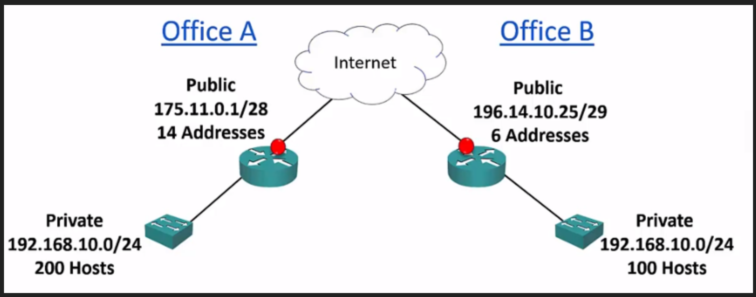
- Given the following topology environment:



- Bank A has a public and a private part of their network. The public one is used to access the internet (175.11.0.0/24). The private LAN cannot access the internet (10.10.10.0/24).
- The same configuration is followed by Bank B. Notice that the public LAN of each bank is different, and that is to be expected, but the private is exactly the same. This is not a problem since these IPs are only significant in each bank's network. So no conflict here. They can't use the same public addresses since they would conflict.

- The IPv4 Global Address Space Problem, the solution IPv6:
  - Let's revisit the problem of the IPv4 design.
  - Remember that the class A loopback address 127.0.0.0 removed millions of ips that could been allocated to organizations? The designers of the ipv4 protocol did not forsee that the IPs would soon run out and the available 4.3 billion addresses would not be enough.
  - They ran out of addresses. The internet authorities started to predict address exhaustion in the late 1980's, and IPv6 was developed in the 90's as the long term solution.
  - IPv6 uses a 128 bit (16 byte) address, compared to IPv4's 32 bit (4 byte) address.
  - IPv6 provides more than 7.9×10^{28} times as many addresses as IPv4. Keep in mind that because 16 bytes and 4 bytes there is an exponential difference, its not just "4 times as large", its actually exponentially bigger (7.9×10^{28} bigger to be exact).
- The IPv6 problem and NAT:
  - There is not a seamless migration path from the IPv4 and IPv6.
  - Changing from ipv4 to ipv6 is not easy, one is not compatible with the other, so until organizations can upgrade from ipv4 to ipv6, a technology (a work around) was developed as a temporary solution to this, its called NAT.
  - NAT stands for Network Address Translation, and it basically works like this:
    * Let's say that we have an organization that has the private IP addresses mentioned in the last example. So it has a public and a private LAN. With NAT the private addresses can communicate with the internet as if they were public. So the organization can use the RFC 1918 private IP addresses which are not publicly routable, so they don't work in the public internet. On the way out, NAT converts the communications coming from the private IPs to public IP addresses.
    * So when then the internet trafic goes out from your private IP address, NAT transforms it to a public IP and the internet can send trafic back through NAT.
    * Many hosts on the inside that are using the RFC 1918 private IP addresses can share a single public IP on the outside.
- Now for an example of private IP addresses and NAT:

– Given the following topology:

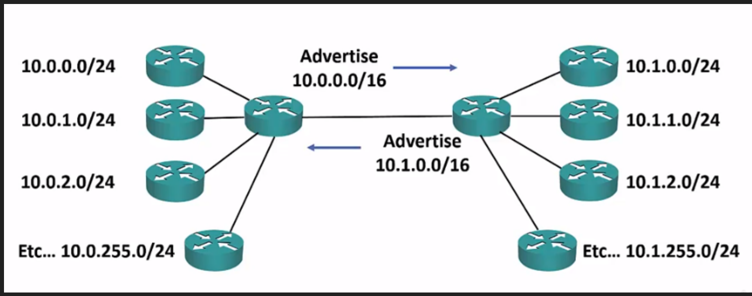


– Office A and B have the same configuration, a range of public and private IP addresses. The private IPs cannot communicate with the internet. Not yet...

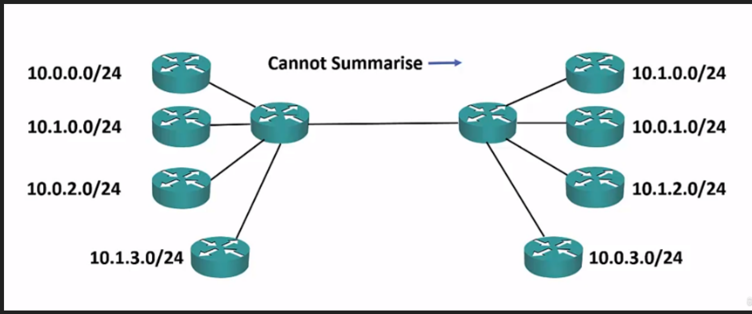
– So NAT will translate the private IPs whenever they are communicating with anybody on the internet. This saves lots of addresses in the ipv4 public address space. And it saves organization's money because public IPs cost money, we can have just one, and have loads of private IPs, and just communicate through that one public IP using NAT.

6.11 Private IP addresses part 2

- Today's networks:
  – Back in the 2000s everybody was saying that they were living the final days of ipv4. But the truth is that it has not happened, almost everybody is using ipv4 with NAT using RFC 1918 private addresses.
  – There are several reasons for this, first there is security benefit of hiding hosts inside a private LAN, the hosts do not have a privately routable IP address. Another reason is that ipv6 has not completely taken over is because most people have more experience using ipv4. Ipv6 is actually a very different format, so most people are not familiar to the format of ipv4.
  – Despite this, ipv6 is still found in some companies, primarily service provider networks, mobile services and large countries that adopted the internet later such as India and China.
  – Spare public IPv4 addresses were exhausted in 2011 so IPv6 is still the future path. It will be a slow process of adoption but eventually ipv4 will be deprecated and ipv6 will completely take hold.
  – Subnetting is used in RFC 1918 private IP addressing to optimise performance and security.
  – You also need to understand subnetting to be able to troubleshoot IP.
  – Because most people use the private IP RFC 1918, it is common to see /24 subnets being used for end hosts, /30 subnetting for point to point links, and /32 for loopback. Remember that if we use the private IPs with NAT, we can have all the range of IPs as if they were ours, so all the possible values from 0.0.0.0 to 255.255.255.255 is ours to do with as we please. We communicate trafic with the internet using our public IP RFC 1918.
  – Using many subnets in the same networks is called VLSM is more common in enterprises which use public IP addresses on their side of networks and need to maximize their use.
- Contiguos addresses and route summarization:
  – Even if you are using private IPs, route summarisation still needs to be used.
  – The folllowing example:



- Notice that the advertise address in the center left router is 10.0.0.0/1y6, this is because it is the common bits that all the IPs have, all of the IPs start with 10.0. Same on the right side, all IPs start with 10.1.
- So when using private IPs you must be careful allocating them, always look to allocate them in contigous blocks. It would be a mistake to not allocate them in contigous blocks like this:



- Notice in the above image that we cannot summarize because the IPs don't have much in common, in this scenario the routers would have to have a list of all routers complete addresses. The IPs dont start with 10.0, but are different i.e. 10.1, 10.0. so you cannot summarise them. Always plan your private IP addressing address space.

6.12 Where to get more subnetting practice?

- Consider using this to study for CCNA.
- `http://www.subnettingquestions.com/`
  - For questions on subnetting.
  - Every refreshing of the site will give you a new question.
- `http://www.subnetting.org/`
- `https://subnettingpractice.com/`
- Search in google for Subnetting calculators and questions.